

MATLAB and Simulink
Robotics Arena



MathWorks Minidrone Workshop *ASEE 2025*

Noah Roberts, *MathWorks*



Noah Roberts

Education Application Engineer

- Worcester Polytechnic Institute (WPI) – 2021
 - Mechanical and Electrical Engineering
- MathWorks – 2022-Present



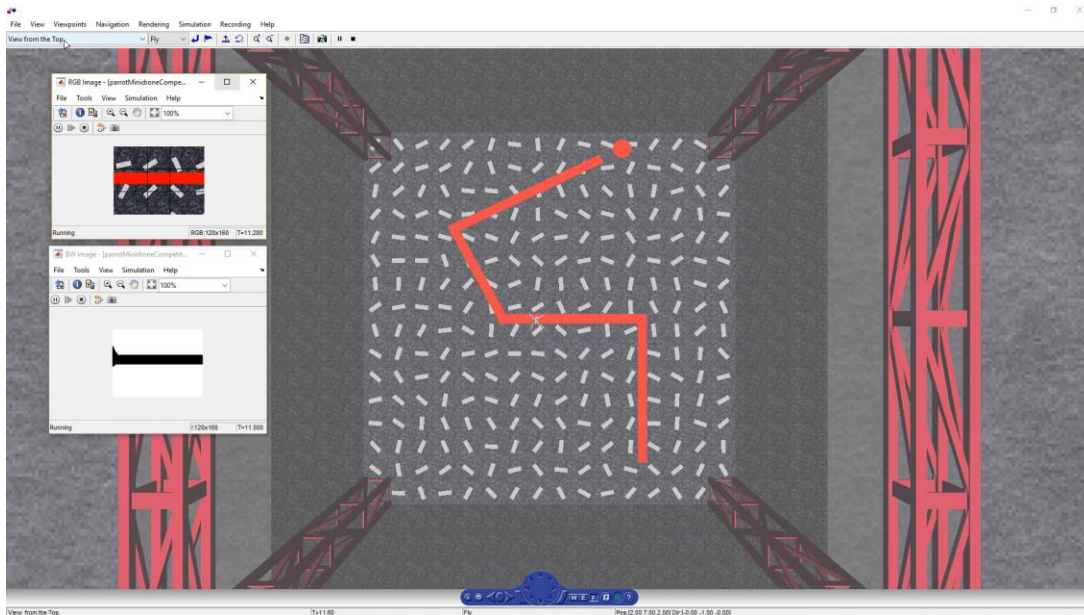
Our Volunteers

Jose Avendano Arbelaez

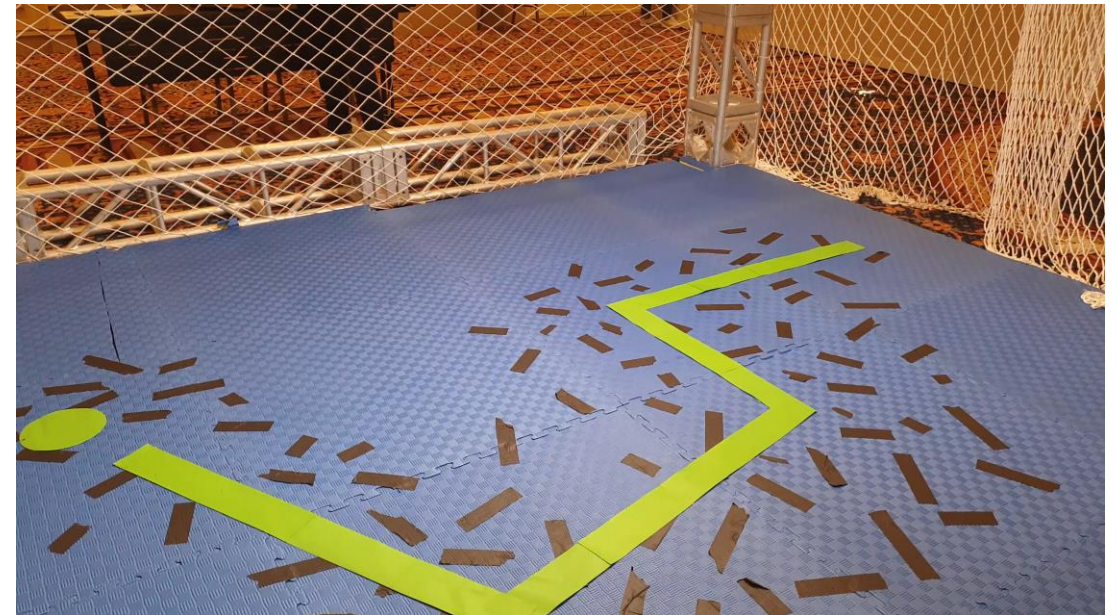
What will we try achieve by the end of the workshop?

Problem Statement: Design an autonomous line follower

Simulation



Real World



Day-wise Agenda

Day 1

START TIME (EST)	END TIME (EST)	Day 1 22 June 2025
9:00	10:00	Welcome & Introduction to workshop
10:00	10:30	Workshop Problem Statement Introduction to Basics of Quadcopters Understanding Model-Based Design
10:30	11:00	<i>Break</i>
11:00	12:30	Path Planning and State Machines Image Processing and Perception
12:30	13:30	<i>Lunch</i>
13:30	15:00	Participants to work on hands-on tasks (Optional) Participants to work on simulations to design an autonomous system (Optional) Testing simulation models on multiple simulation tracks Preview of Day 2 and 3
15:00	15:30	<i>Break</i>
15:30	16:00	Hardware deployment workflow
		<i>Day 1 end</i>



Day-wise Agenda

Day 2

START TIME (EST)	END TIME (EST)	Day 2 23 June 2025
9:00	9:15	Welcome back and recap
9:15	11:00	Working on simulations
		Practice Round 1
11:00	11:15	Team 1
11:15	11:30	Team 2
11:30	11:45	Team 3
11:45	12:00	Team 4
12:00	13:00	<i>Lunch Break</i>
		Practice Round 1
13:00	13:15	Team 5
13:15	13:30	Team 6
13:30	13:45	Team 7
13:45	14:00	Team 8
		Practice Round 2
14:00	14:15	Team 1
14:15	14:30	Team 2
14:30	14:45	Team 3
14:45	15:00	Team 4
15:00	15:15	Team 5
15:15	15:30	Team 6
15:30	15:45	Team 7
15:45	16:00	Team 8
		<i>Day End</i>



Day-wise Agenda

Day 3

START TIME (EST)	END TIME (EST)	Day 3 24 June 2025
9:00	9:15	Welcome back and recap
9:15	10:00	Hover the drone to test the hardware (5 mins per team)
		Optional Practice Round 3
10:00	10:15	Team 1
10:15	10:30	Team 2
10:30	10:45	Team 3
10:45	11:00	Team 4
11:00	11:15	Team 5
11:15	11:30	Team 6
11:30	11:45	Team 7
11:45	12:00	Team 8
12:00	13:00	<i>Lunch Break</i>
		Final Competition
13:00	13:15	Team 1
13:15	13:30	Team 2
13:30	13:45	Team 3
13:45	14:00	Team 4
14:00	14:15	Team 5
14:15	14:30	Team 6
14:30	14:45	Team 7
14:45	15:00	Team 8
15:00	15:30	<i>Certificates, Interaction, feedback, and giveaways</i>
		<i>Day End</i>



Pre-requisites

MATLAB Onramp

Learn the essentials of MATLAB® through this free, two-hour introductory tutorial on commonly used features and workflows.

[Details and launch](#)

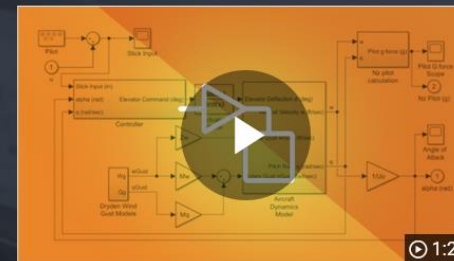


Simulink Onramp

Learn the basics of how to create, edit, and simulate models in Simulink® with this free, three hour introductory tutorial.

Prerequisites: [MATLAB Onramp](#)

[Details and launch](#)



Pre-requisites

- MATLAB R2024b Installed
- Support Package Installed



Simulink Support Package for Parrot Minidrones

by MathWorks Simulink Team **STAFF**

Design, simulate and deploy algorithms to fly Parrot Minidrones (Supported from 23b)

[+ Follow](#)

- Check if Toolboxes Installed
 - **<https://tinyurl.com/38jnvx3y>**
 - Download and run **toolboxCheck.m**



Let's rearrange

Team Formation

1

2

3

4

5

6

7

8

[Question]

How many of you have worked on drones/aerial vehicles?



Agenda

- Basics of Quadcopters
- Understanding Model-Based Design
- Path Planning
- Perception
- Designing your Autonomous System



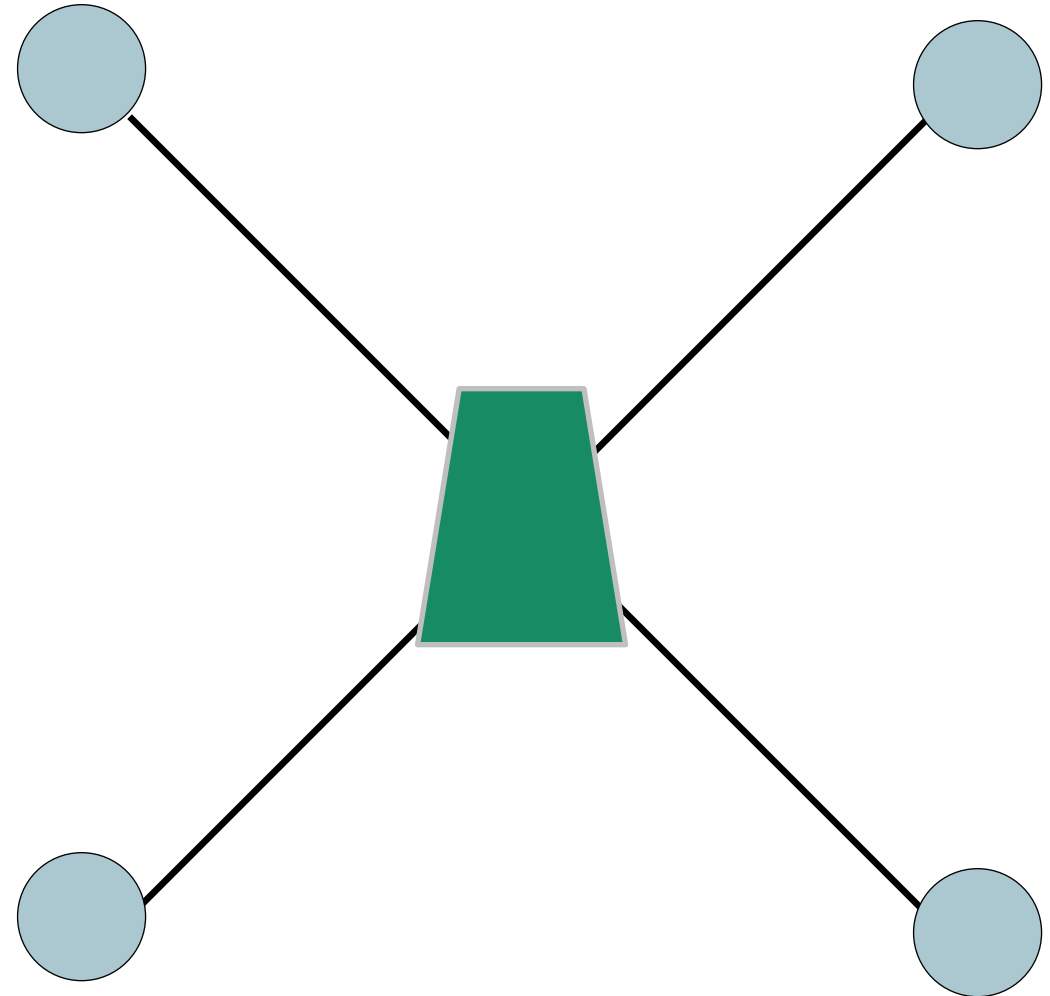
Agenda

- Basics of Quadcopters
- Understanding Model-Based Design
- Path Planning
- Perception
- Designing our Autonomous System



What is a quadcopter?

- Rotating wing aircraft (rotorcraft)
- 4 rotating propellers
- Rotors used to generate lift



Knowing our quadcopter system

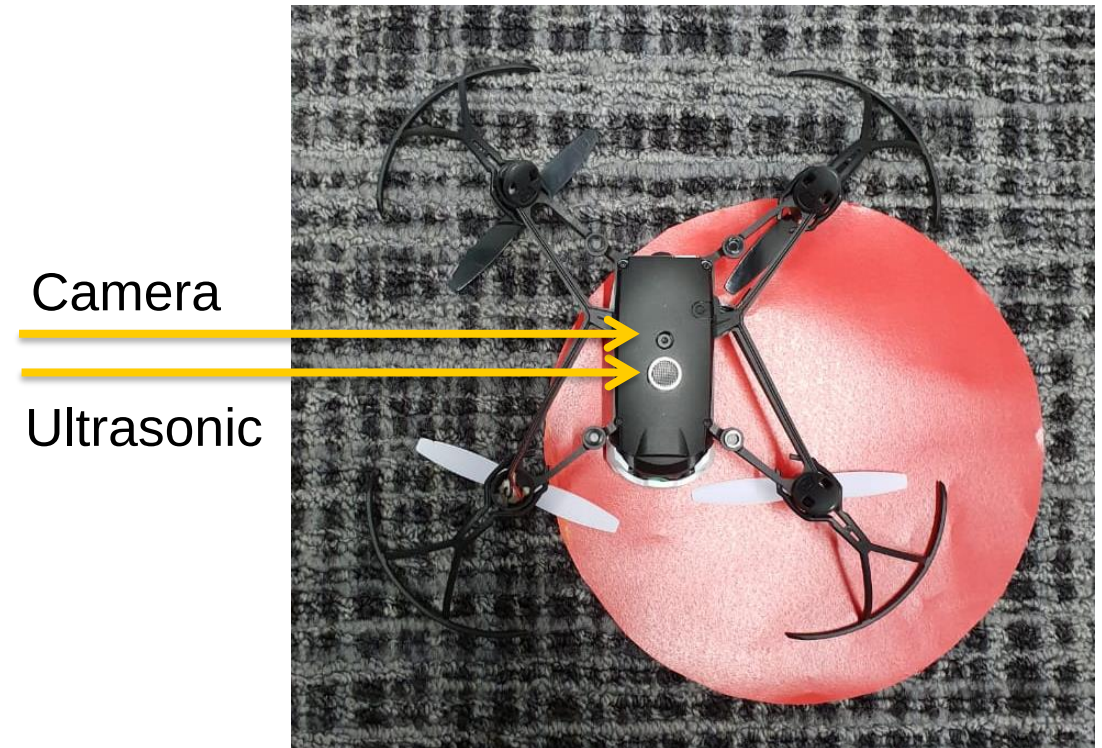
- Work with the already available hardware
 - Parrot Mambo Minidrone
- Sensors
- Actuators
 - 4 motors each with its propeller



Knowing our quadcopter system

Sensors

- Ultrasonic Sensor
 - To calculate altitude
- Camera
 - To capture the view
 - To calculate horizontal velocities
- Pressure Sensor
 - To calculate altitude
- Inertial Measurement Unit (IMU)
 - 3-axis accelerometer
 - 3-axis gyroscope



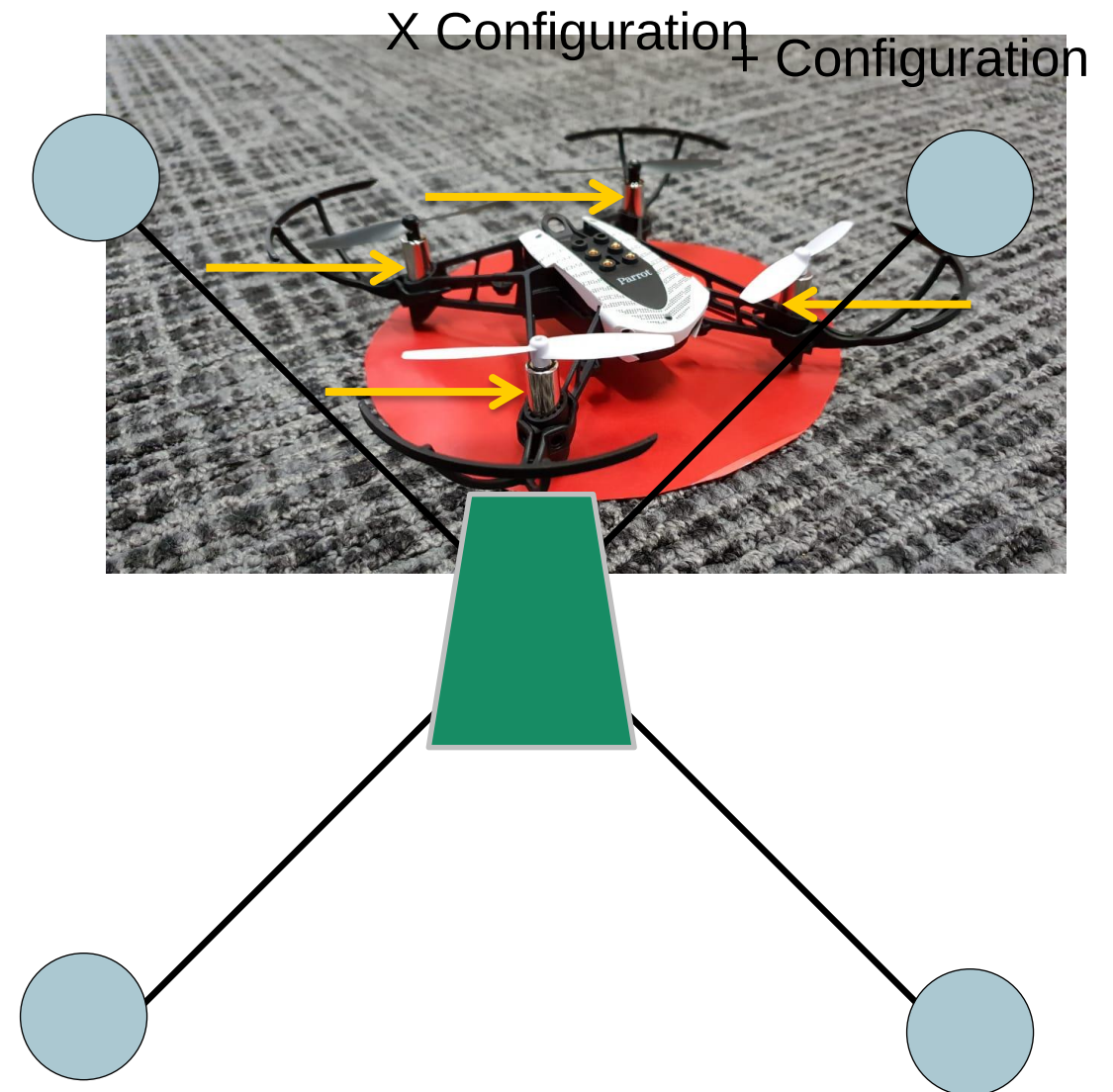
What is the configuration of the Parrot Mambo?



Knowing our quadcopter system

Actuators

- 4 motors
- Provide motor commands
 - Command drone to move in specific directions
 - Decide which motor to send command to and what value to send
- Configuration:
 - X configuration
 - + configuration



Equations of motion



Work on the concept of balancing forces and torques

Translational Dynamics:

$$F_{\text{frame}} = F_{\text{gravity}} - F_{\text{thrust}} - F_{\text{drag}}$$

Rotational Dynamics:

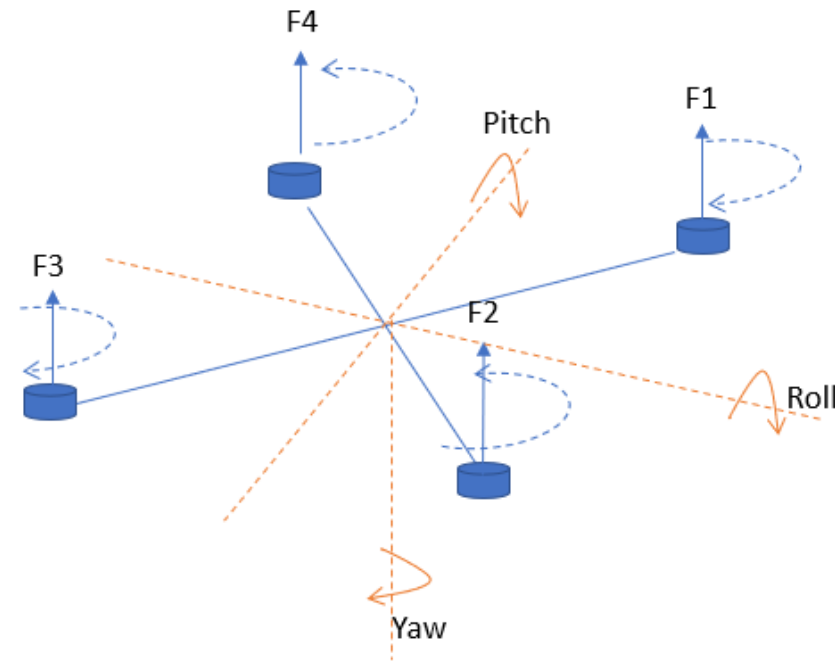
$$\tau_{\text{frame}} = \tau_{\text{motors}} - \tau_{\text{gyro}} - \tau_{\text{inertia}}$$



Drone motions

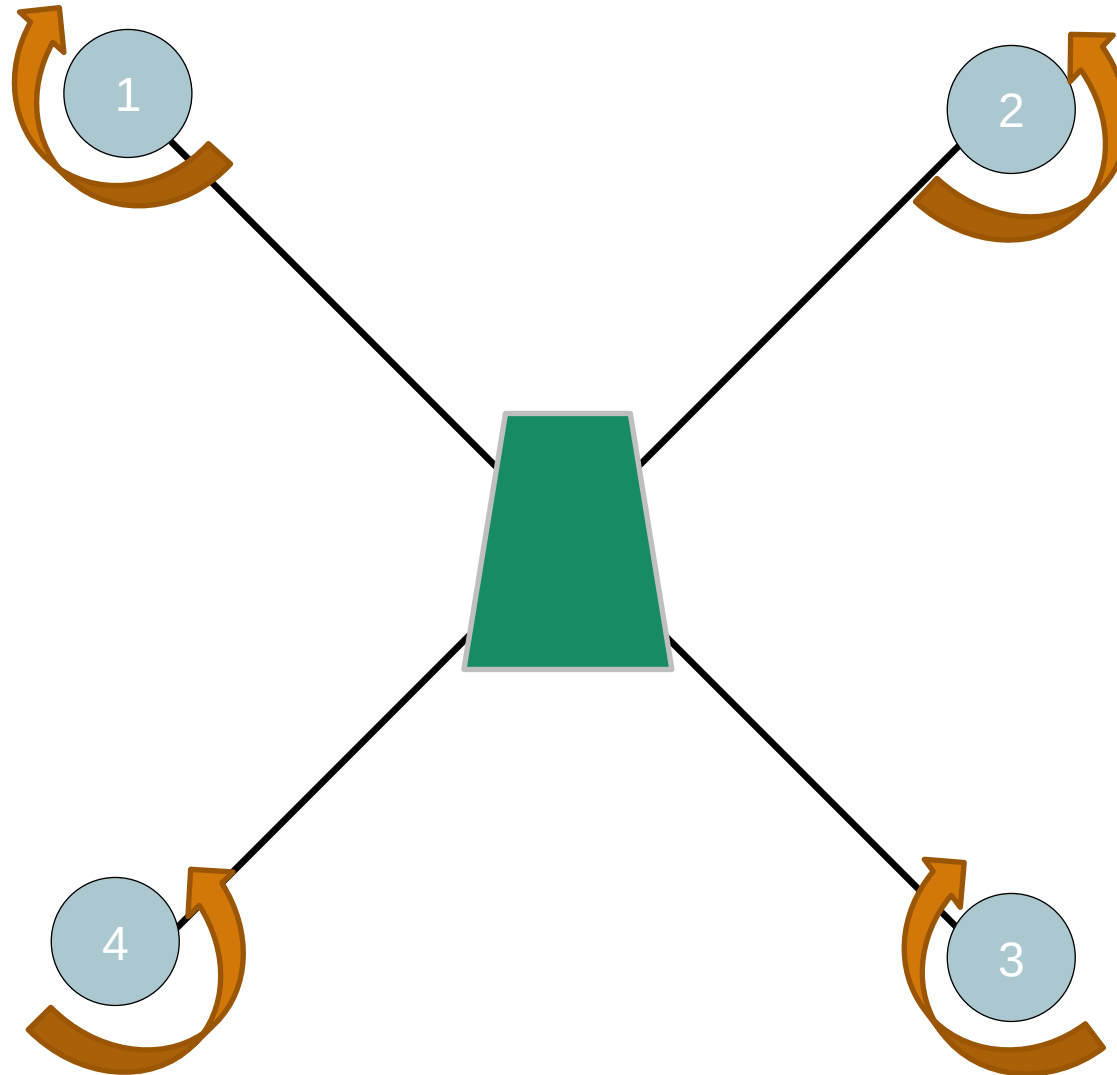
- 6 degrees of freedom
 1. Up-down
 2. Left-right
 3. Forward-backward
 4. Rotation around X-axis: Roll
 5. Rotation around Y-axis: Pitch
 6. Rotation around Z-axis: Yaw

- What will we control?
 - Thrust, Pitch, Roll, Yaw



How do we control the drone the motion?

Thrust

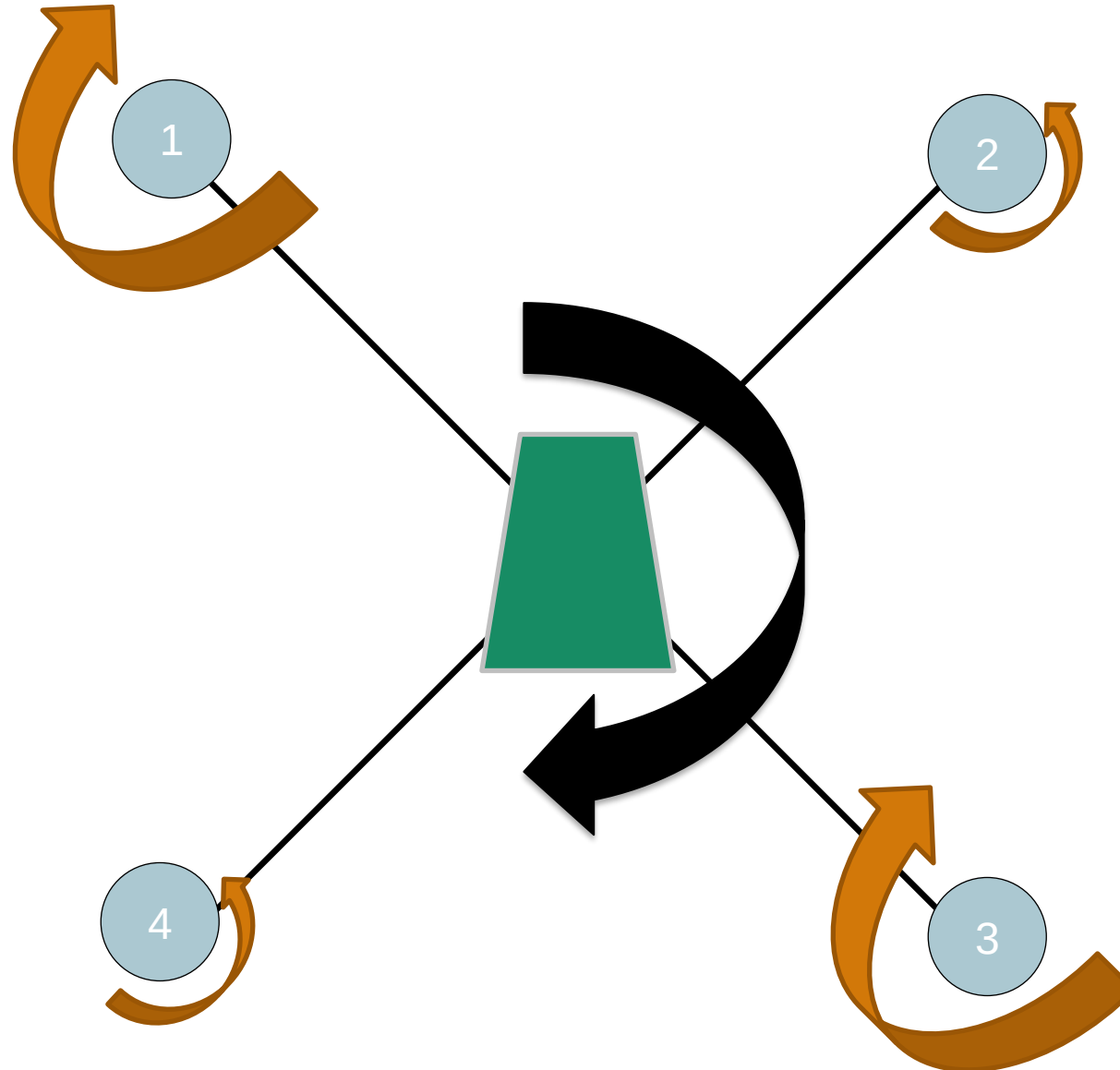


*Motor arrows indicate
motor torque direction*



How do we control the drone the motion?

Yaw

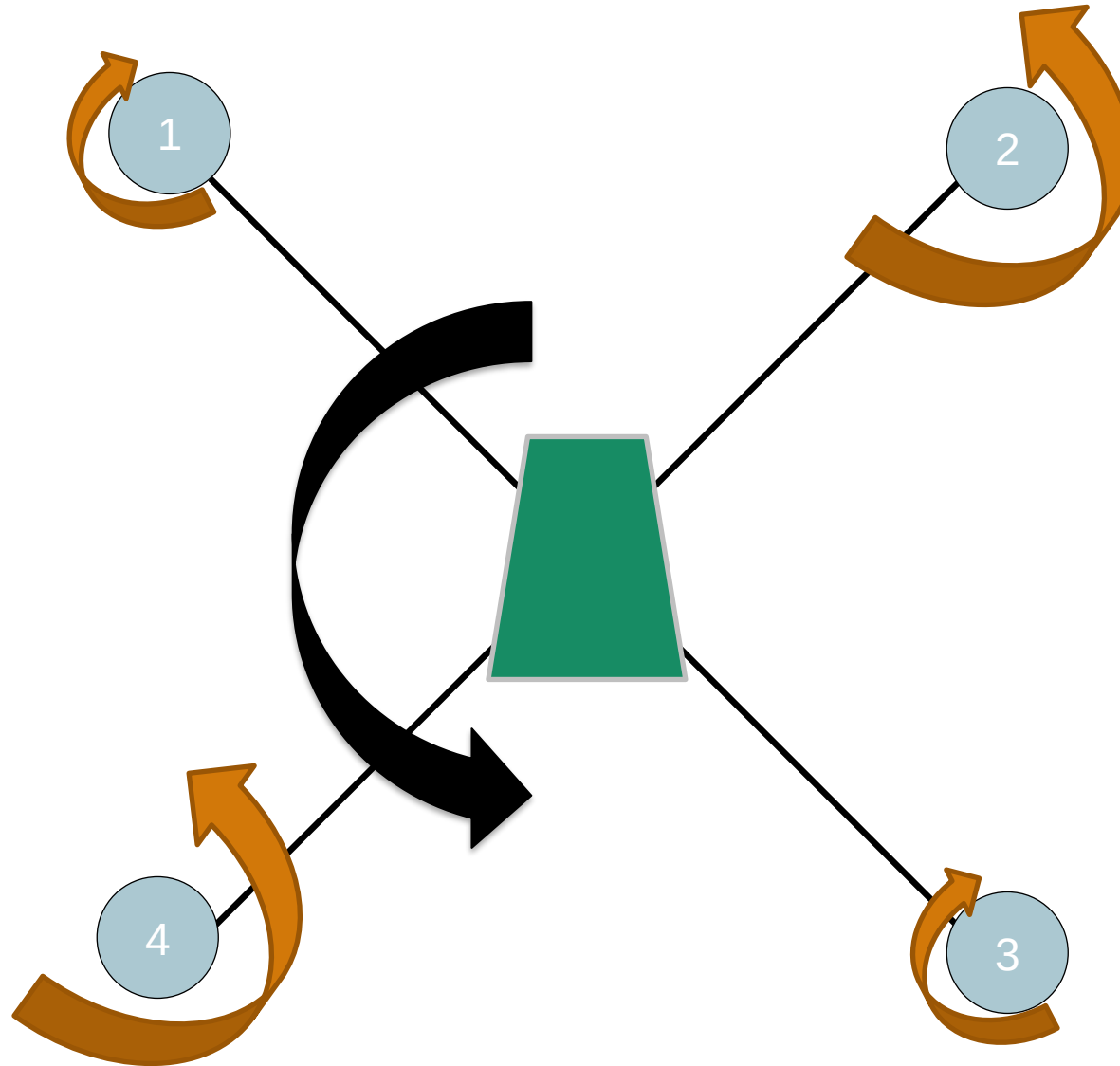


*Motor arrows indicate
motor torque direction*



How do we control the drone the motion?

Yaw

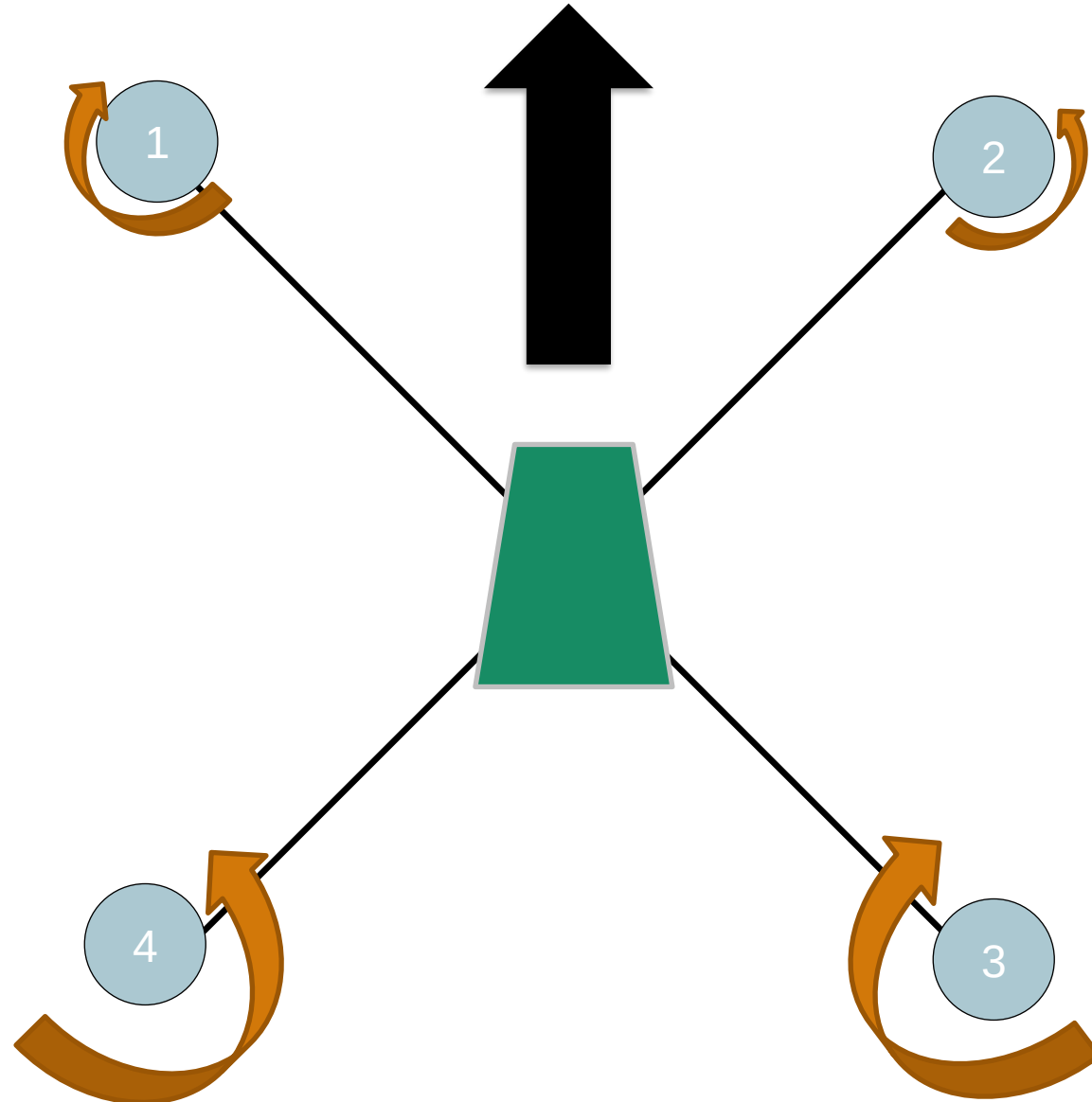


*Motor arrows indicate
motor torque direction*



How do we control the drone the motion?

Pitch

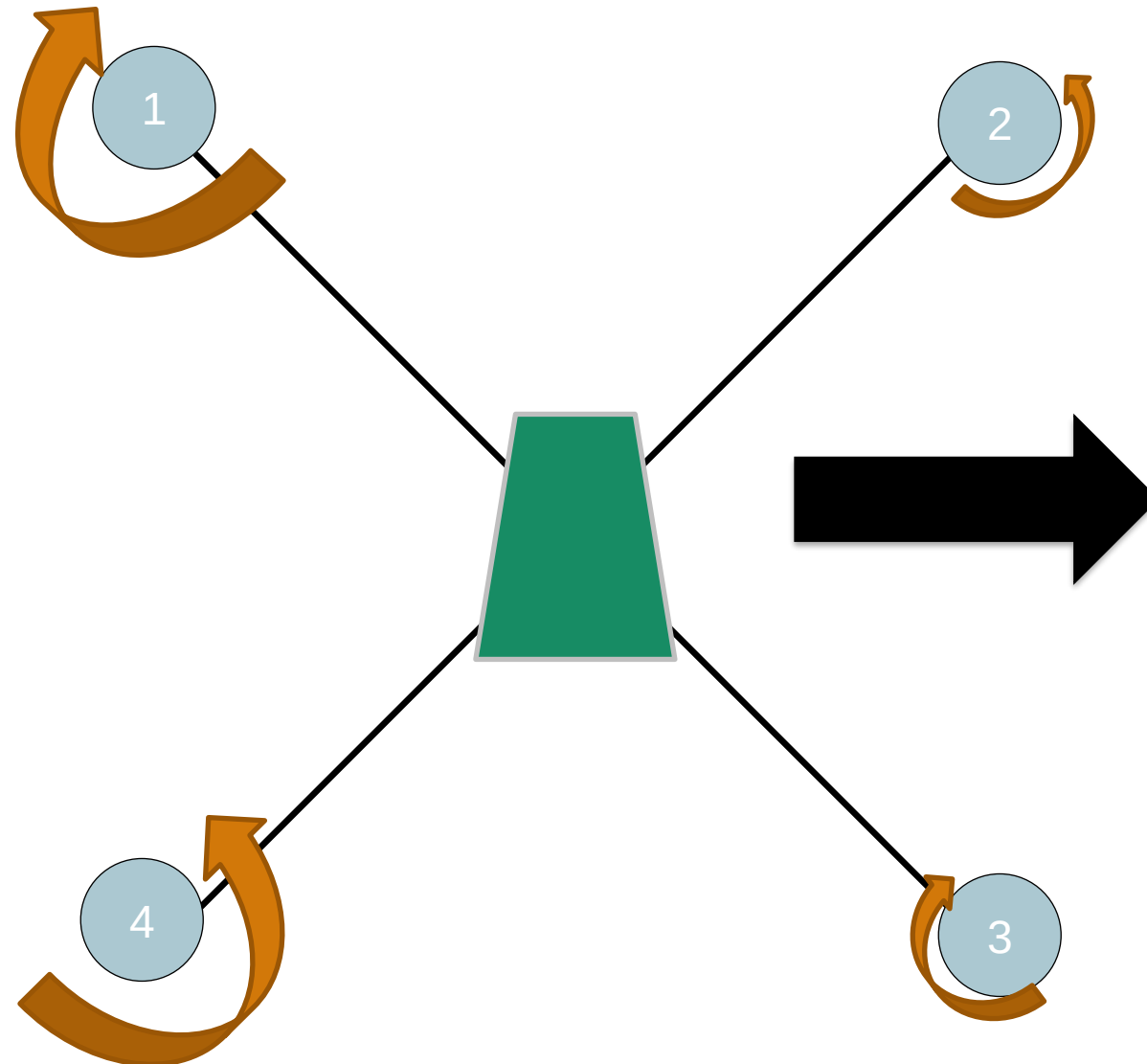


*Motor arrows indicate
motor torque direction*



How do we control the drone the motion?

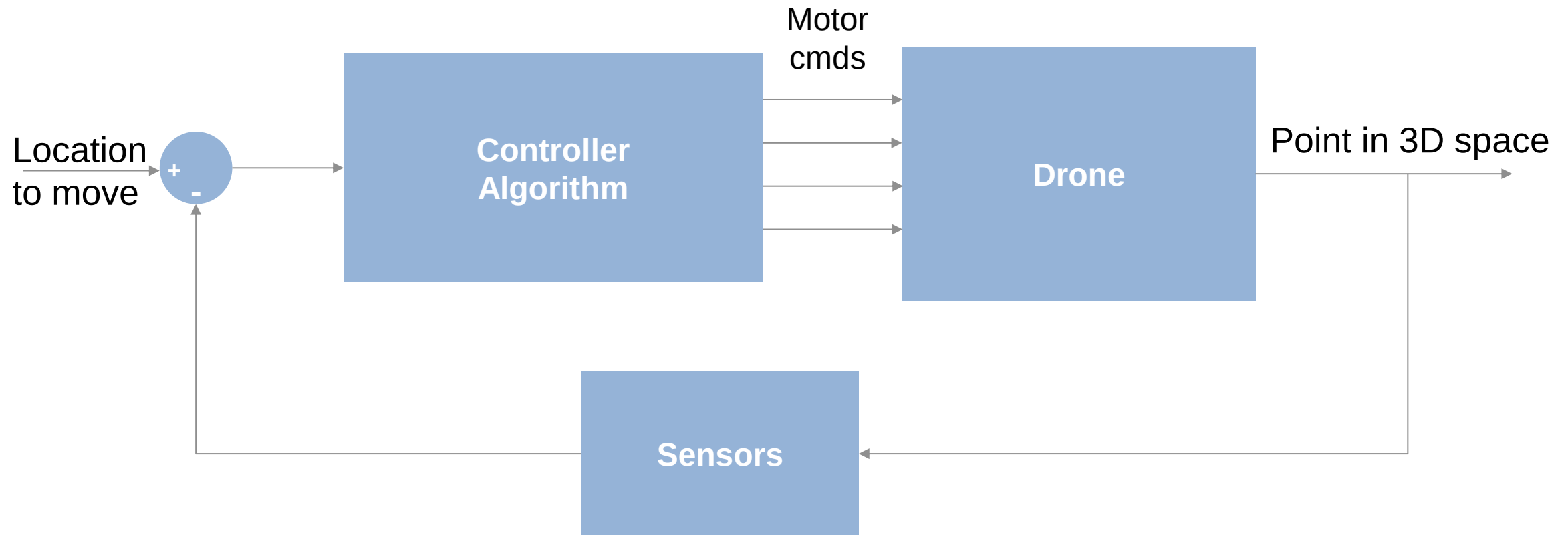
Roll



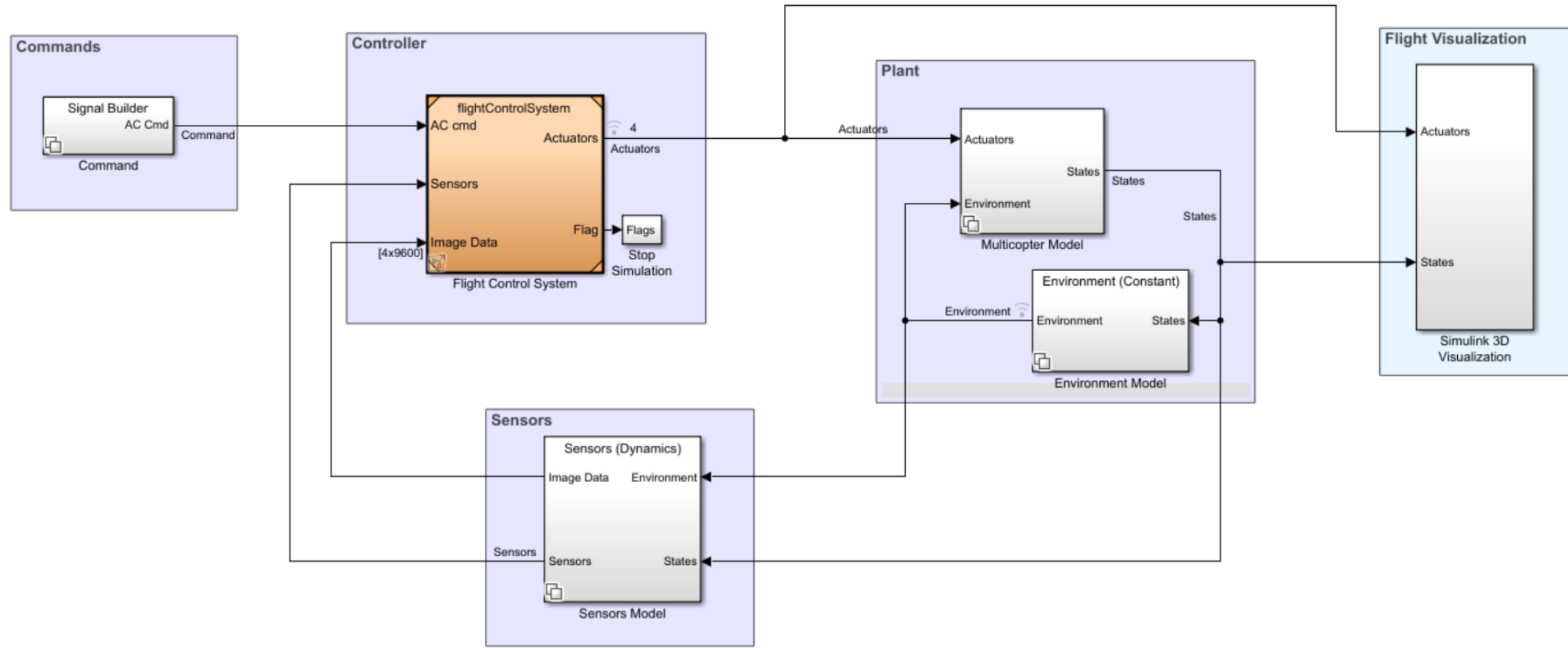
*Motor arrows indicate
motor torque direction*



Modelling a drone



Modelling a drone

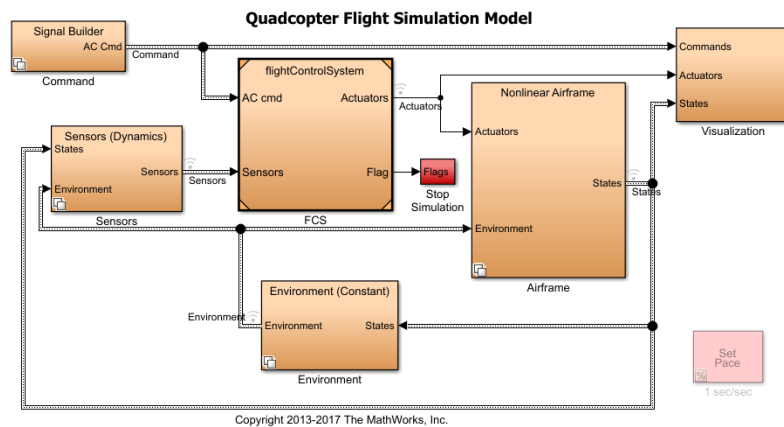
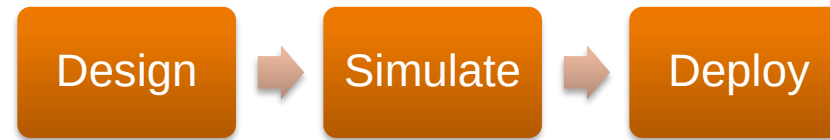


Agenda

- Basics of Quadcopters
- **Understanding Model-Based Design**
- Path Planning
- Perception
- Designing our Autonomous System



Understanding Model-Based Design



SIMULINK®

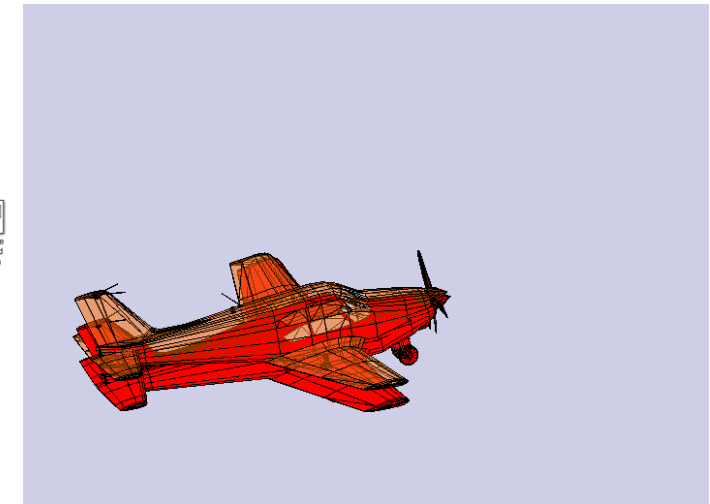
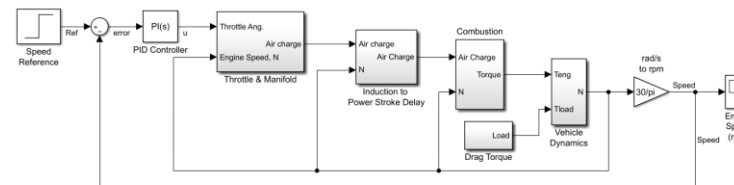
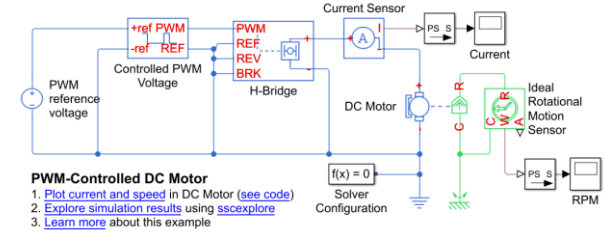
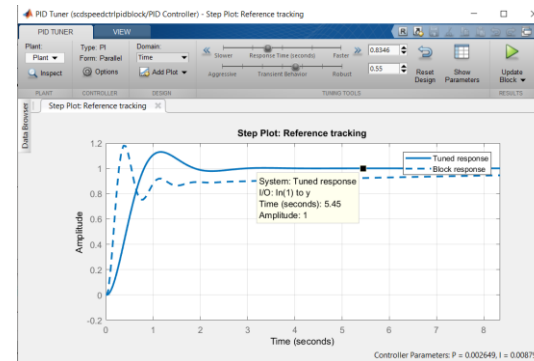
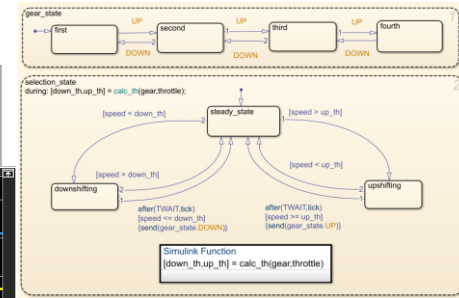
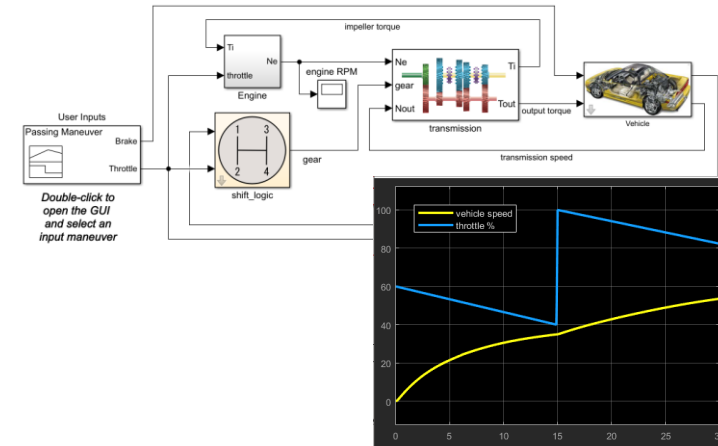
Modeling, simulation, and embedded systems

- Platform for **Model-Based Design**

- **Block diagram** modeling
- **Simulation** of physical systems
- Automatic **code generation**

- Applications in:**

- Control systems
- Signal processing
- Communications systems



Industry Example / Airbus

Commercial Aircraft

Saved three months of development time on the A380, the world's largest passenger jet



Industry Example / Blue Origin

Reusable Rockets

Space travel for private passengers

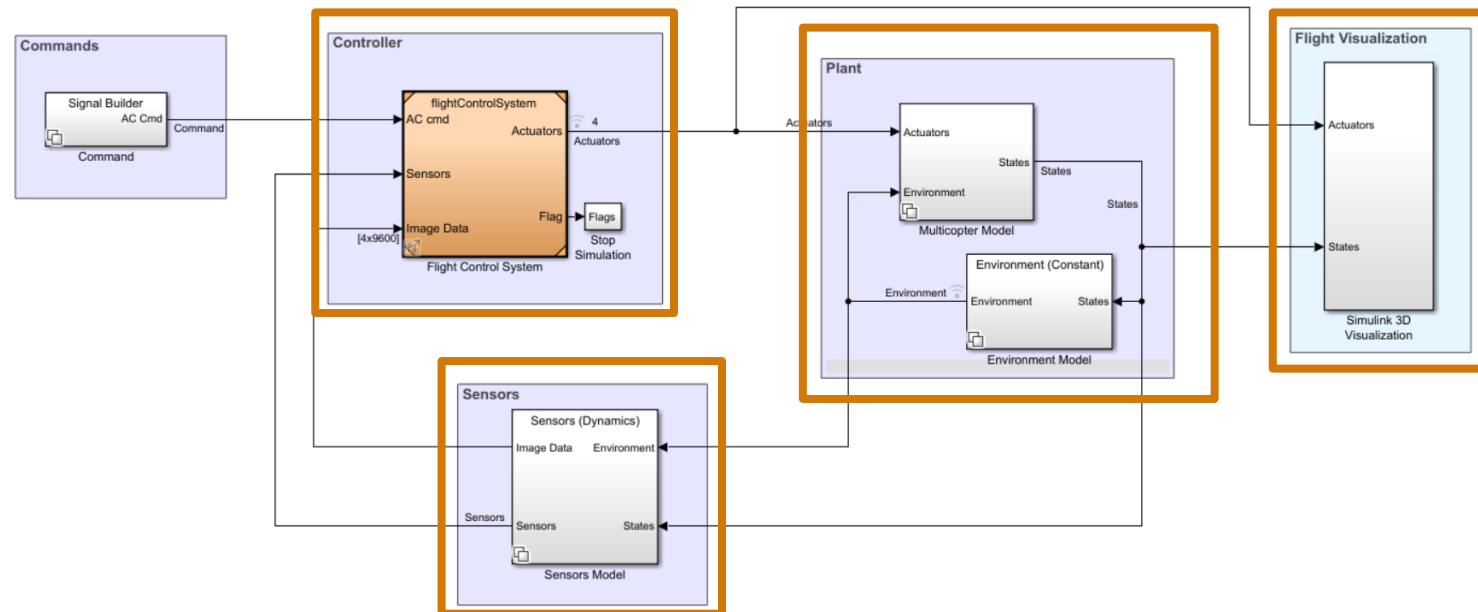
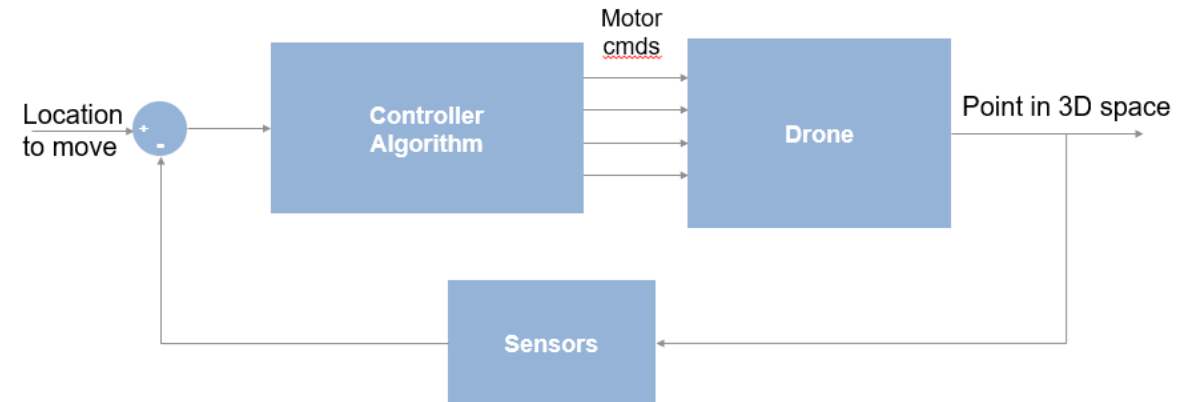
First rocket to enter space and return to Earth with a “soft” landing



Using Simulink Blocks

What do we have in the model?

- Model of a Parrot Mambo Minidrone
 - Plant
 - Sensors
- Visualization environment
- Design of a controller that hovers a drone



Coffee Break

Have a team name !

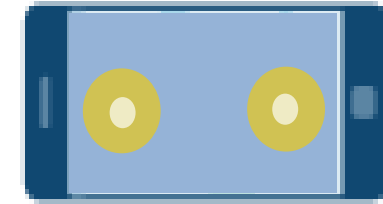


Controlling a drone

Manual control

Parameters controlled independently:

1. Altitude
2. Pitch
3. Roll
4. Yaw



Commands



Drone



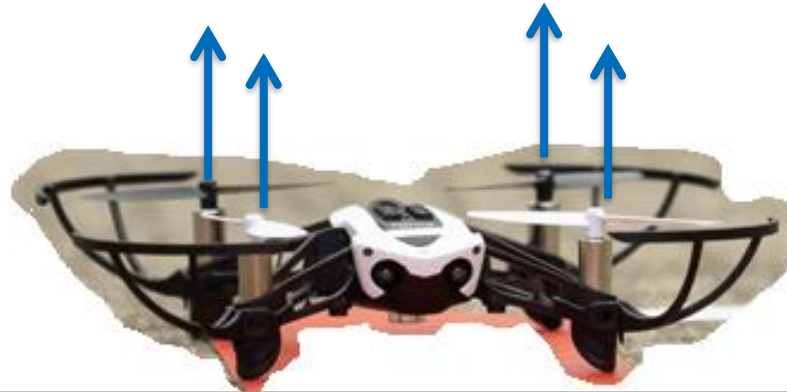
Controller



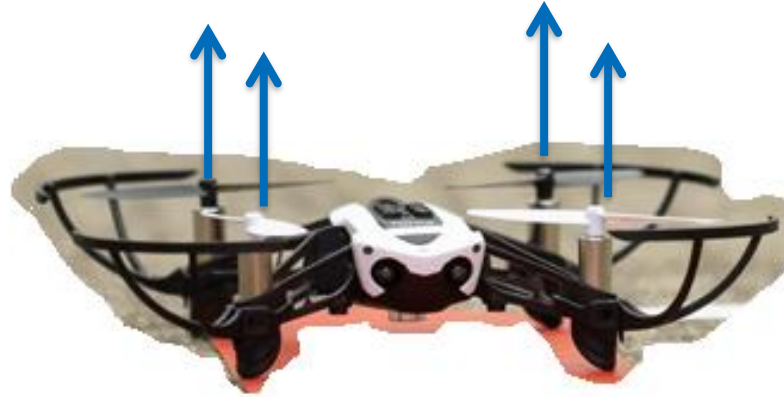


Assume drone is parallel to the floor

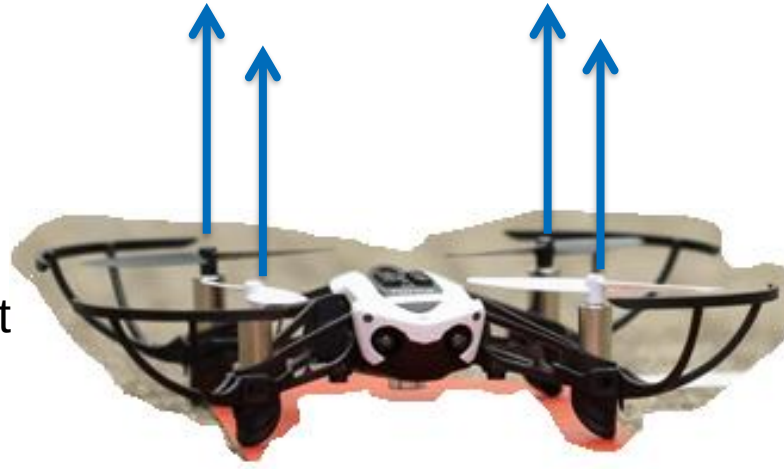




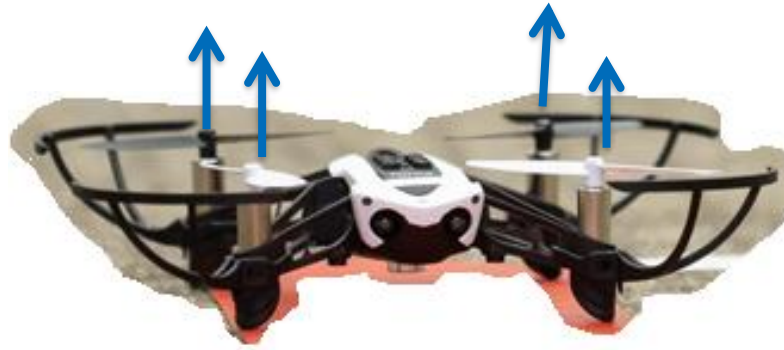
Drone overcomes the gravity force



Increasing thrust -> Increased height



Decreasing thrust -> Decreased height



[Hands-on]
>> parrotMinidroneCompetitionStart



[Hands-on] Parrot Minidrone Competition Model

Image Processing System

Control System

What do these blocks do?

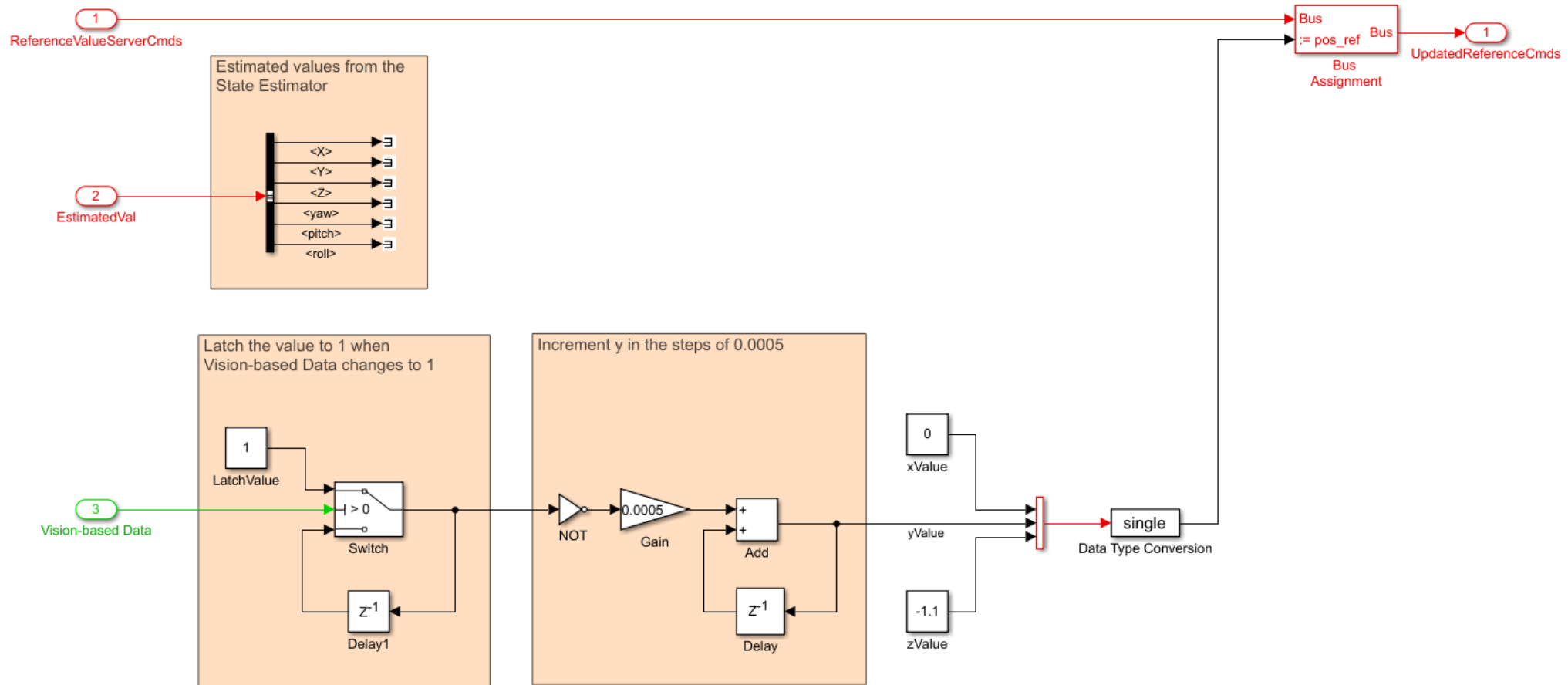


Agenda

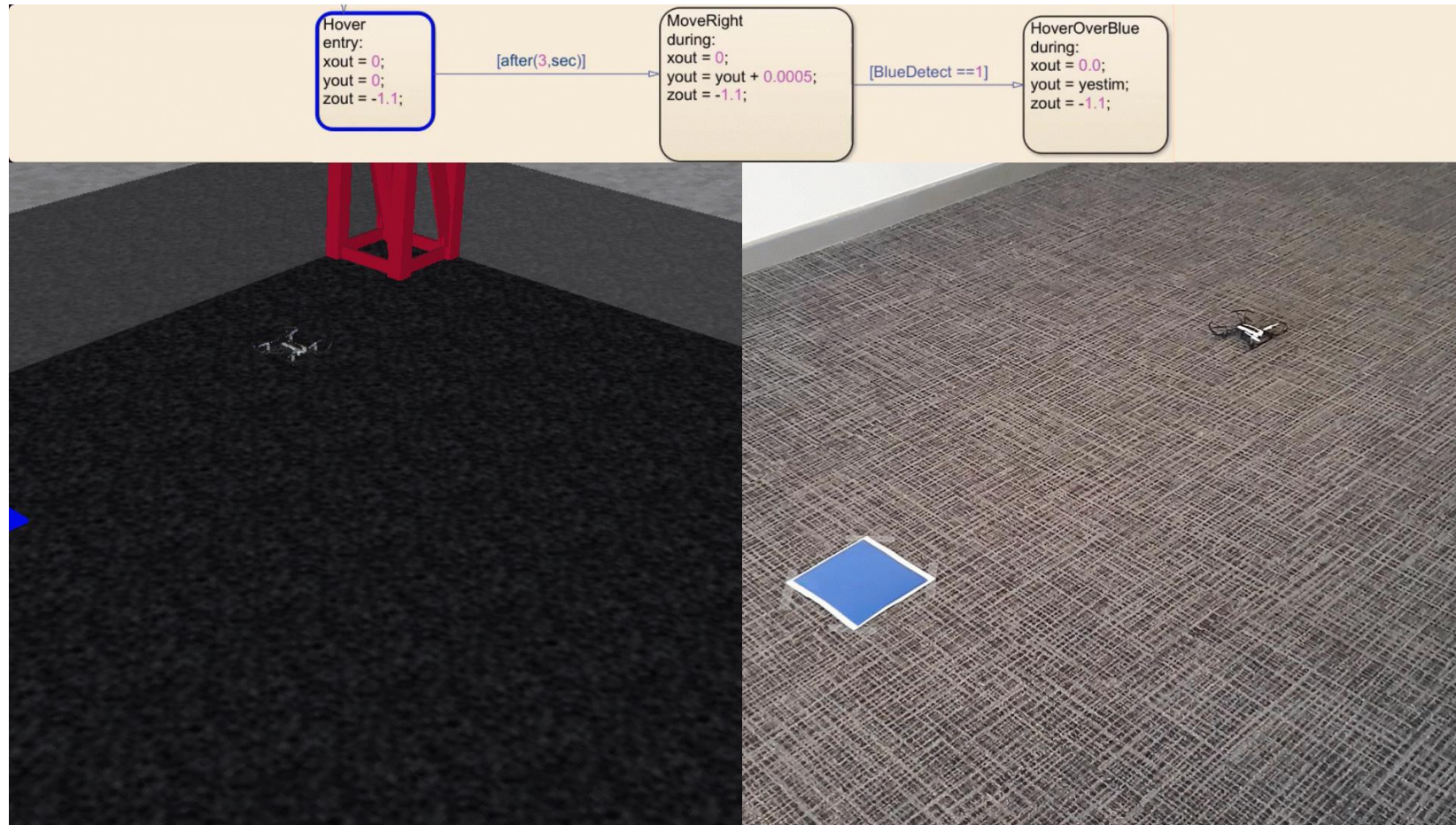
- Basics of Quadcopters
- Understanding Model-Based Design
- **Path Planning**
- Perception
- Designing our Autonomous System



Path Planning

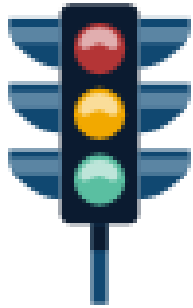


Path Planning



Understanding State Machines

Let's start with a simple example

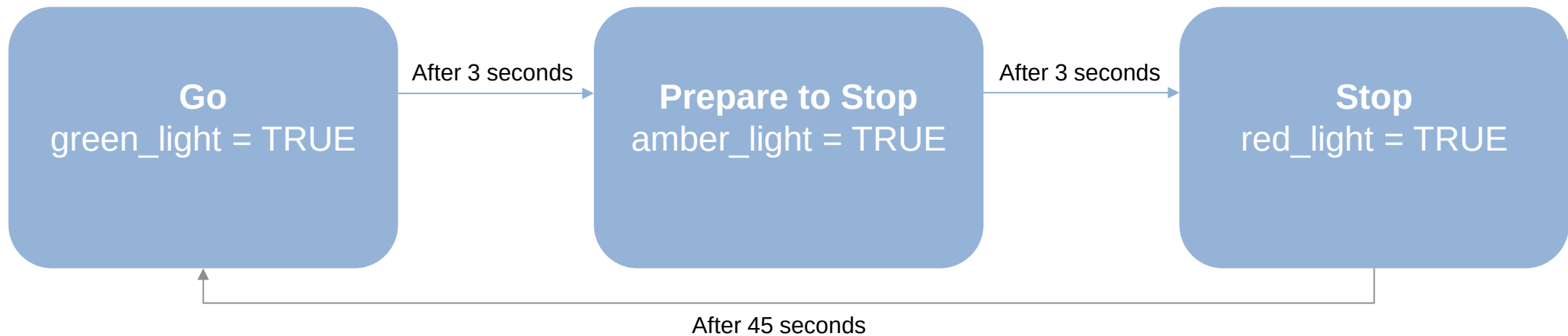


- Represent reactive systems that have states or modes
- States change based on defined conditions and events



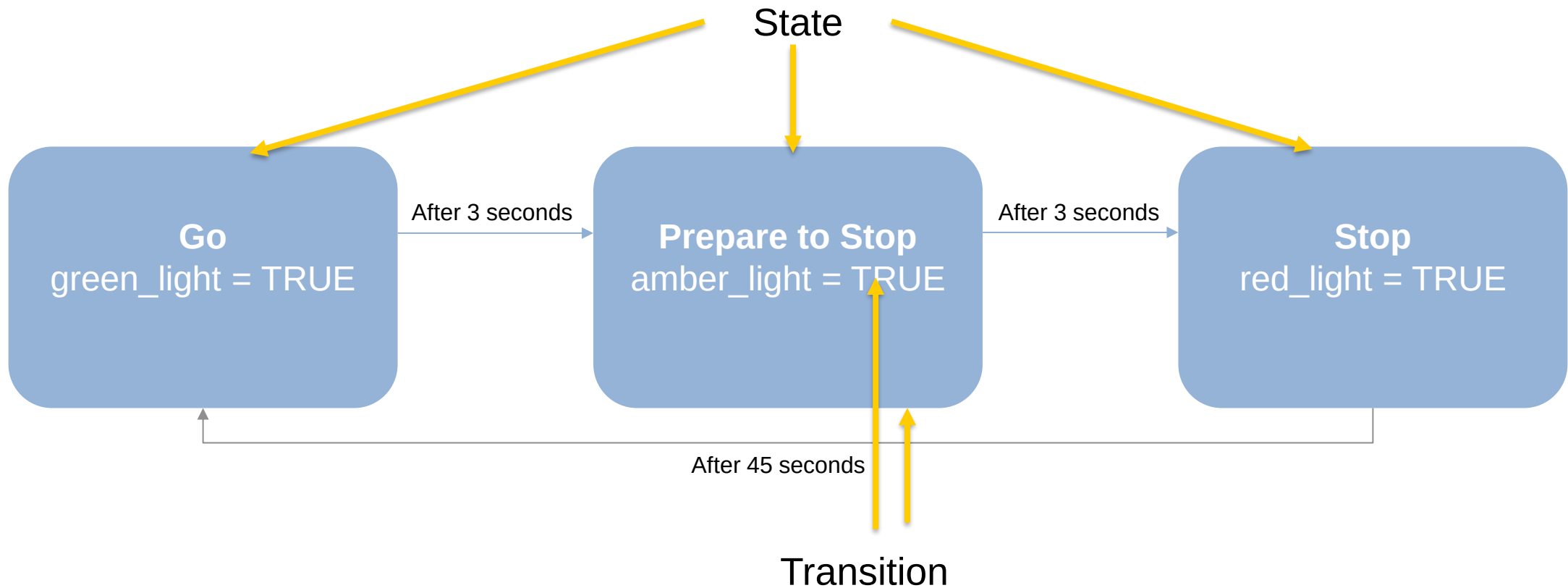
Understanding State Machines

Modeling States and Transitions



Understanding State Machines

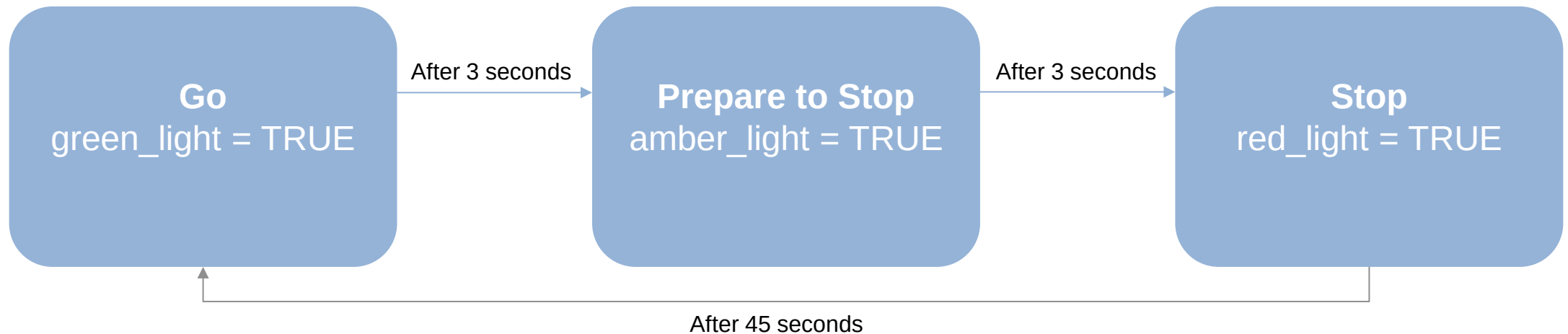
Modeling States and Transitions



Understanding State Machines

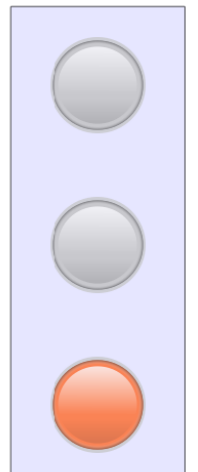
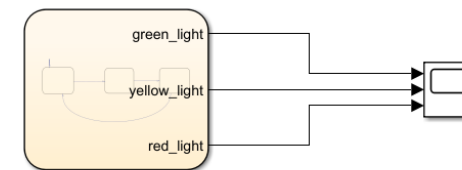
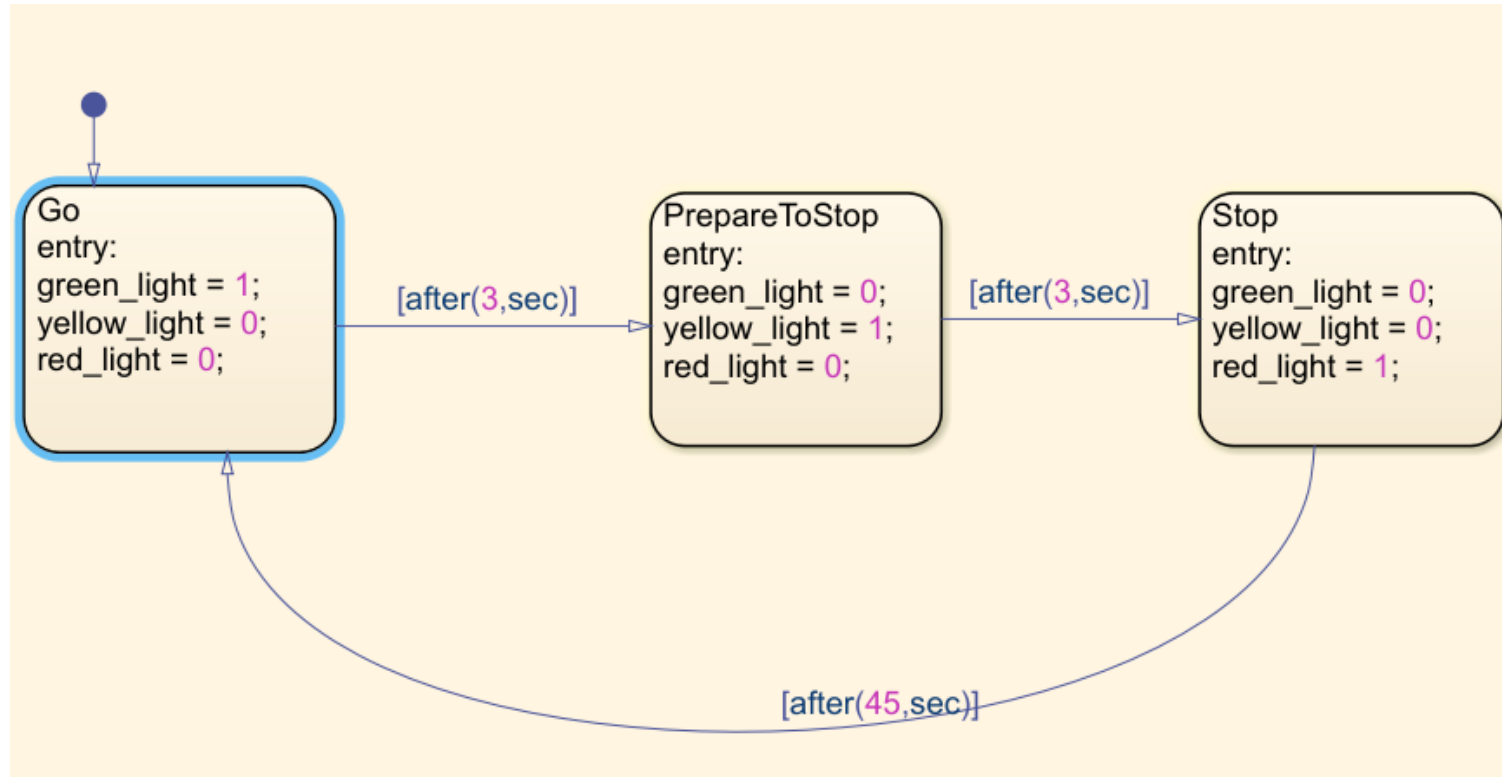
Modeling States and Transitions

- Rules to transition between states
- State transition diagram
- Current state depends on variables and previous state



Let's build a traffic light system together!
>> simulink





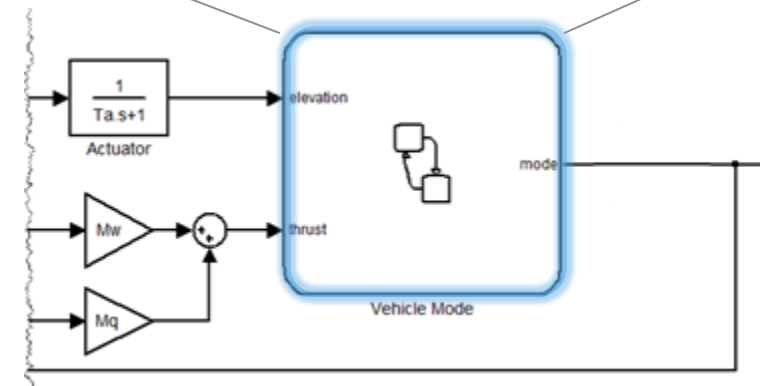
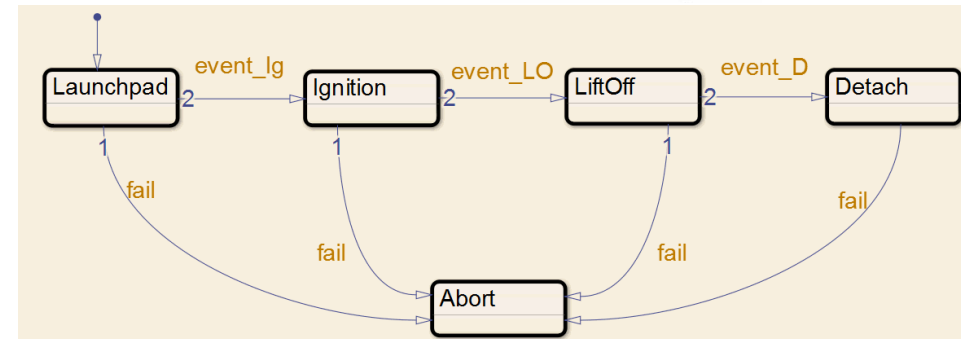
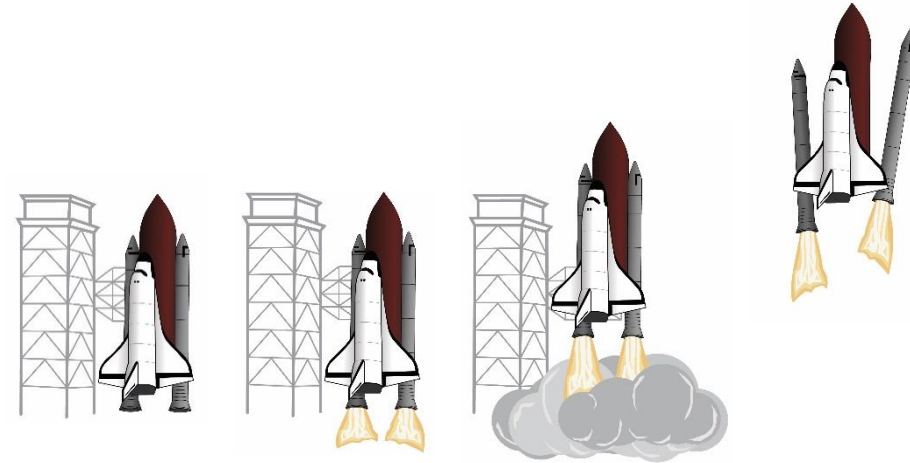
[Question]

What is a State Machine? When can it be used for our autonomous system design?



What is a State Machine?

- Model and simulate decision logic for reactive systems
- Develop mode-logic using state machines and flow charts
- See how the logic behaves with diagram animation and integrated debugger



How is Stateflow different from Simulink?

■ Simulink

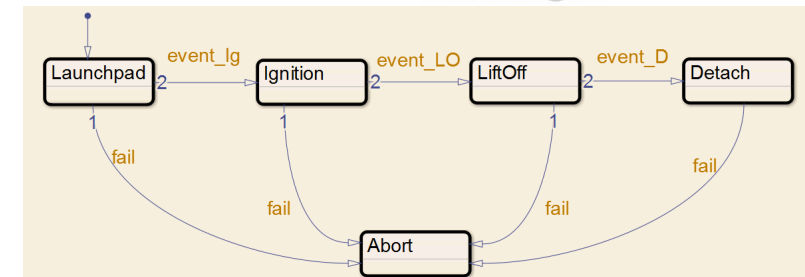
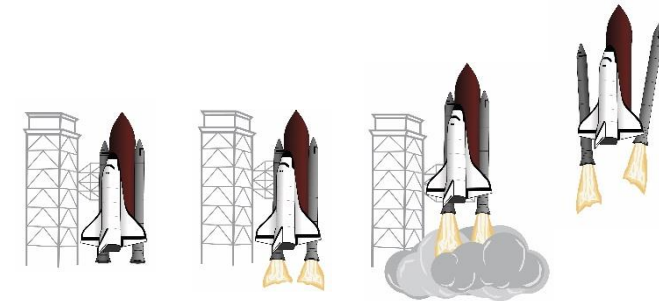
Simulink is used to respond to **continuous** changes in dynamic systems.



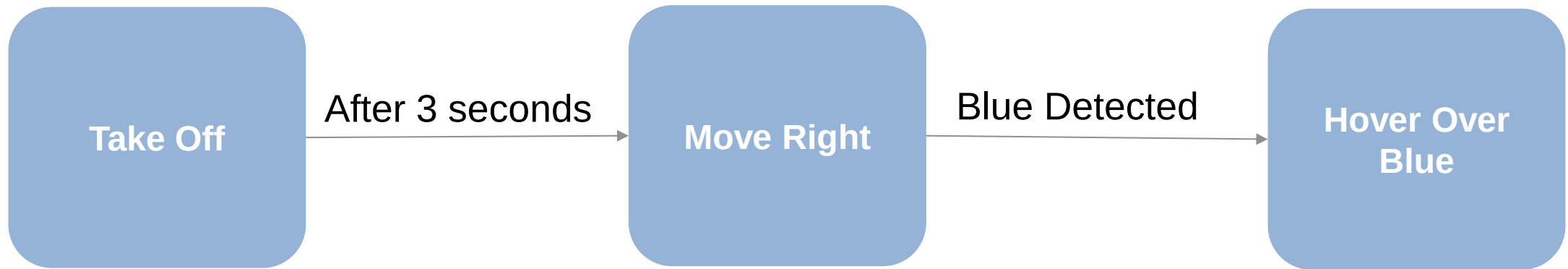
Real-world systems have to respond to both continuous and instantaneous changes.

■ Stateflow

Stateflow is used to respond to **instantaneous** changes in dynamic systems.



Drone Flight States

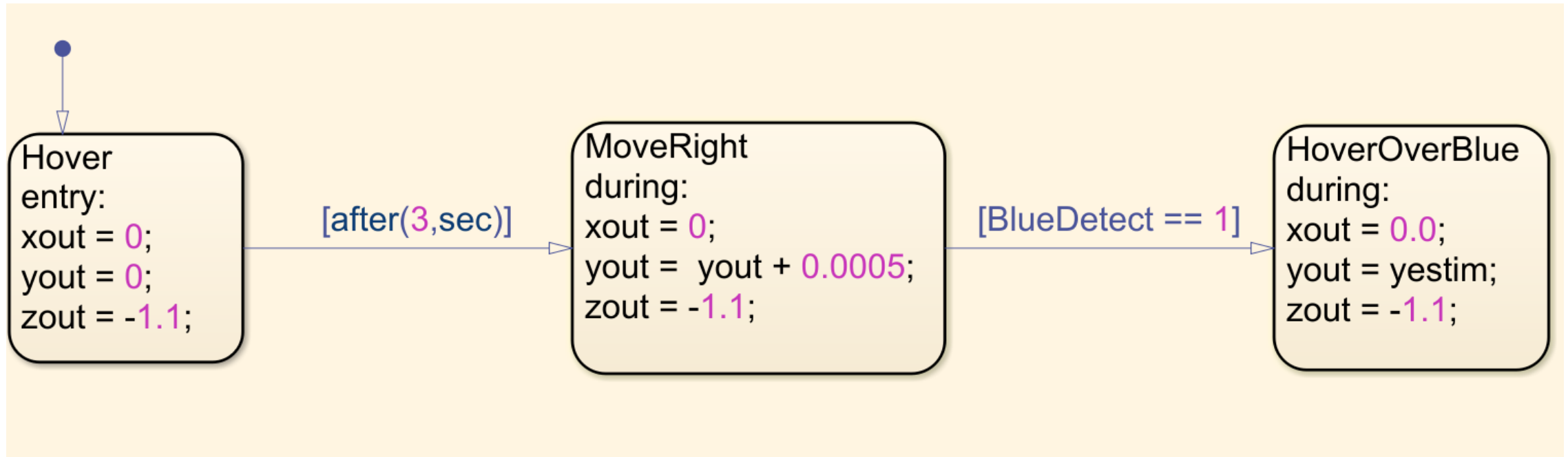


[Hands on] Parrot Minidrone Competition model



Drone Flight States

Stateflow



[Question]

How can we make the drone to follow the entire track?



Agenda

- Basics of Quadcopters
- Understanding Model-Based Design
- Path Planning
- **Perception**
- Designing your Autonomous System

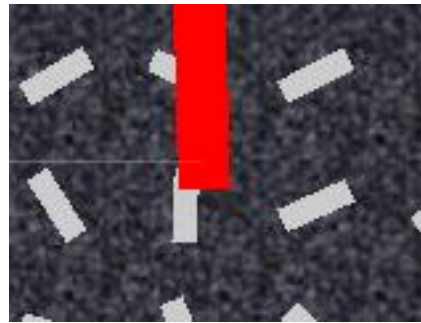


Camera Sensor

- Camera
 - To capture the view
 - To calculate horizontal velocities
- 120*160 image Y1UY2V format image



Real World Camera View



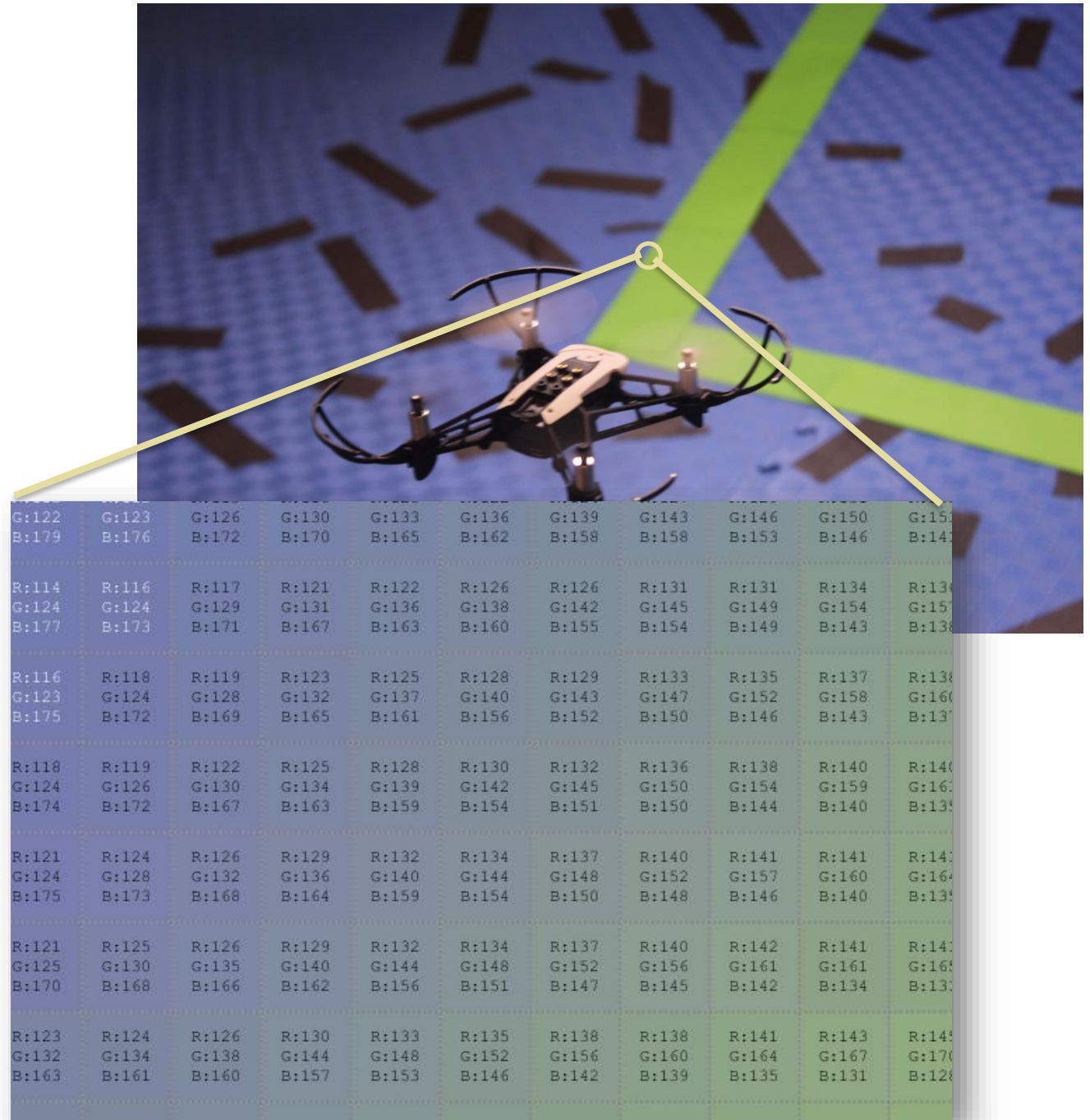
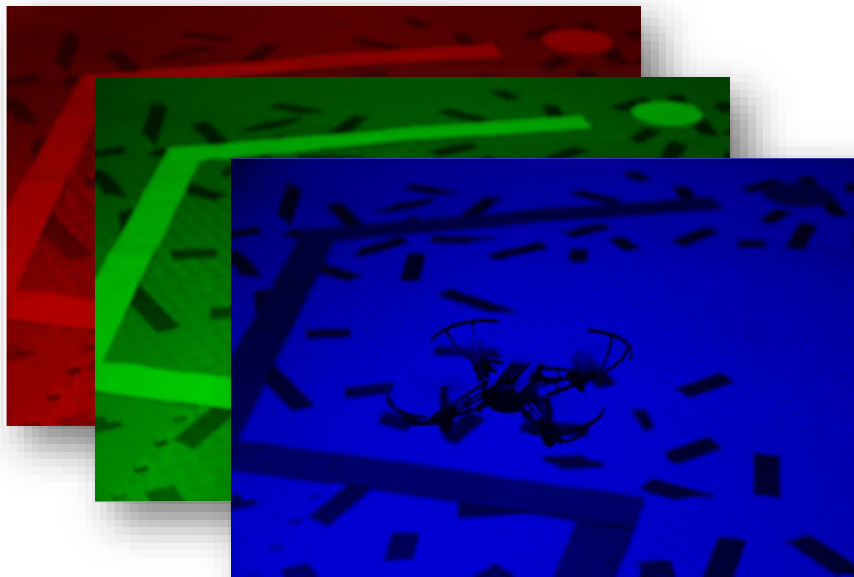
Simulation Camera View

Camera



What is an image?

- Collection of pixels
 - Typically, a pixel has a value between 0 to 255
- Concatenation of channels in third dimension

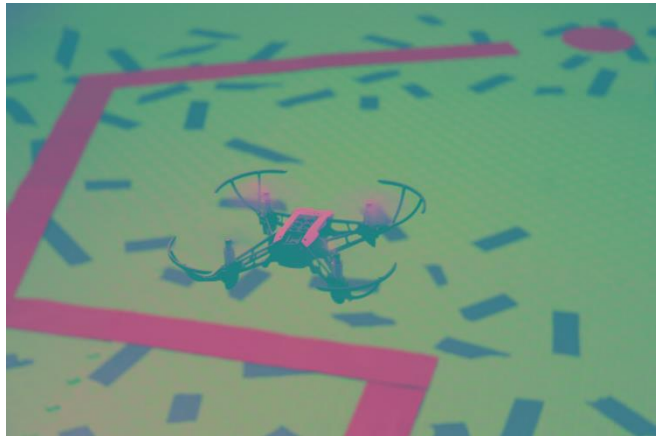


Color Spaces

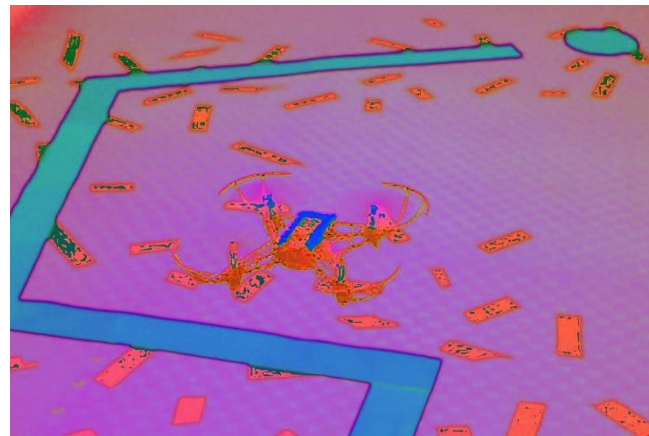
RGB



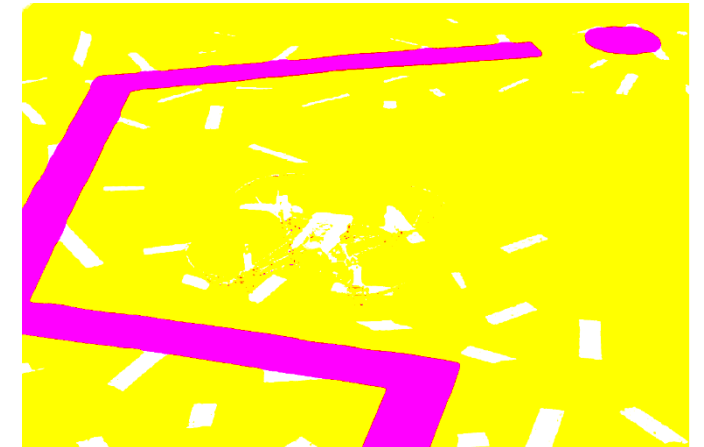
YCbCr



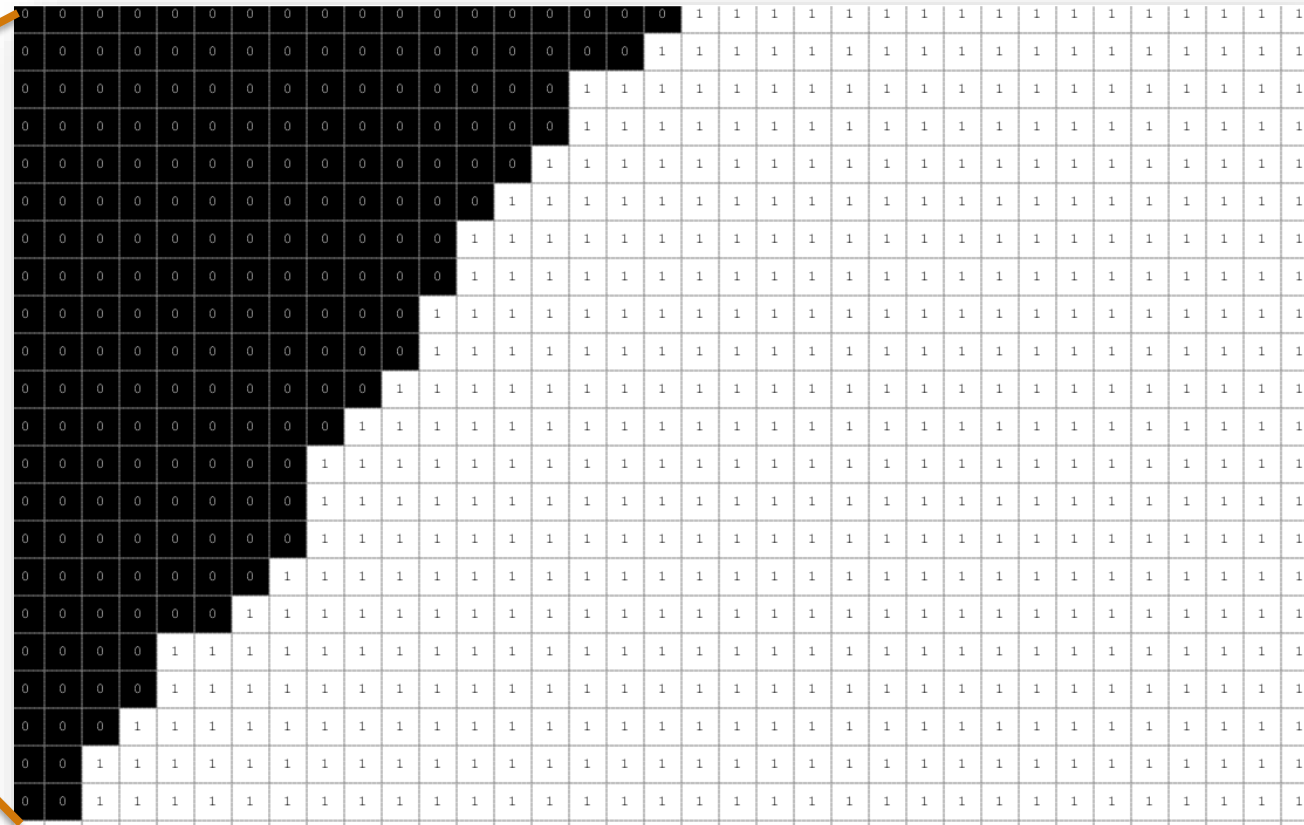
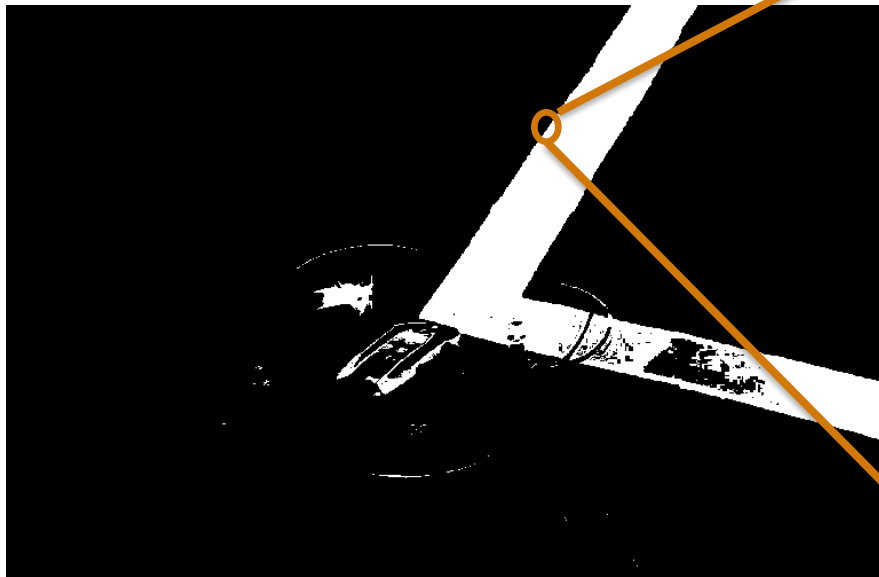
HSV



L*a*b*



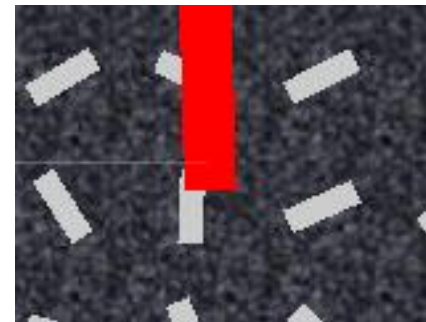
Binary Image



[Hands-on] Color Thresholder Example



Real World Camera View



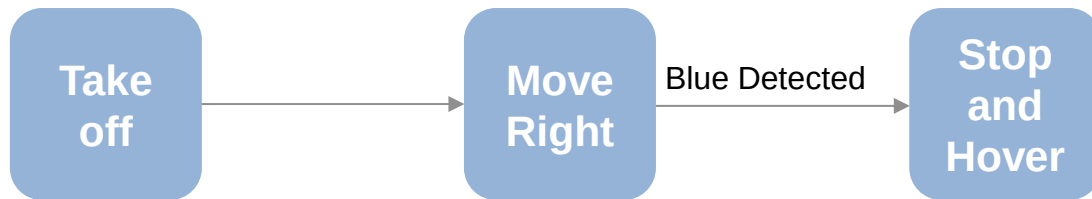
Simulation Camera View



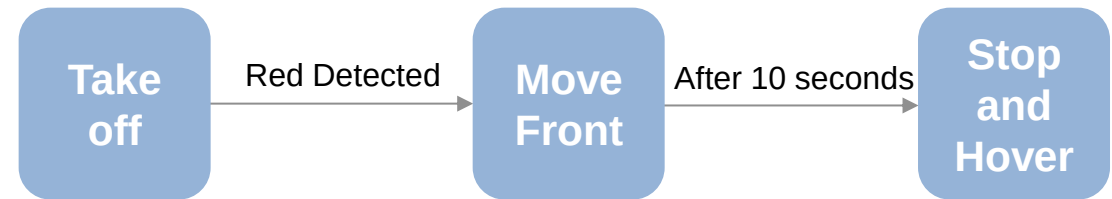
Tracking the first line segment

Task Difference

What we presently do?



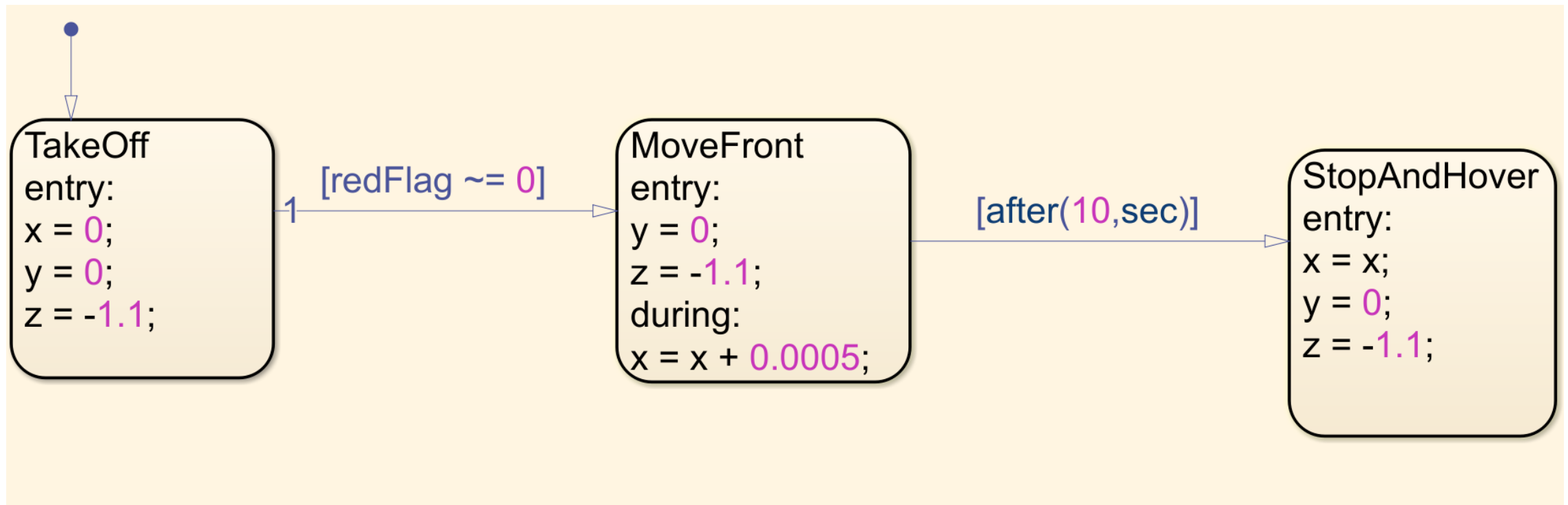
What we want to do?



Note: We are tracking only the first line segment for 10 seconds.

Tracking the first line segment

Stateflow



[Question]
What are the pitfalls in this algorithm?



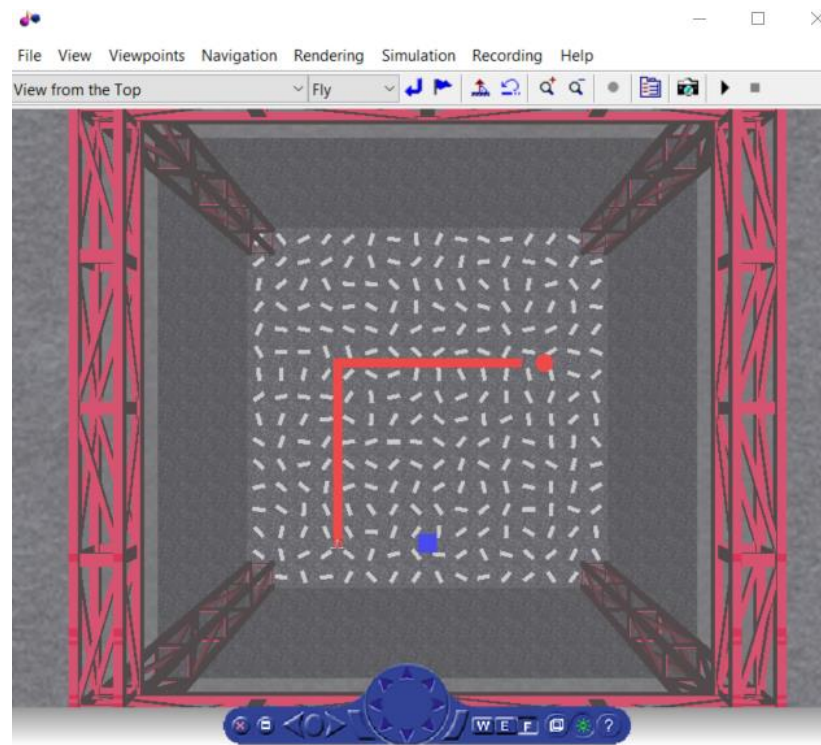
Agenda

- Basics of Quadcopters
- Understanding Model-Based Design
- Path Planning
- Perception
- **Designing your Autonomous System**



What will we try achieve by the end of the workshop?

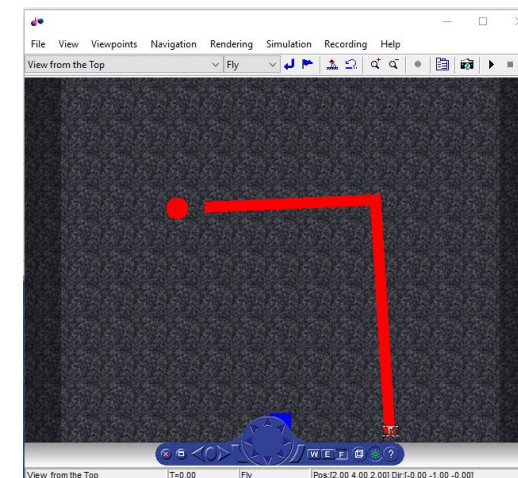
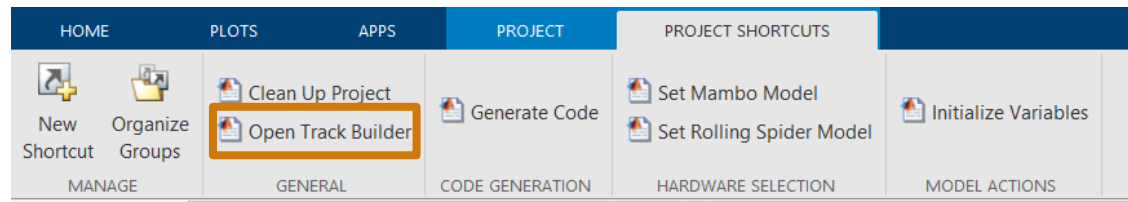
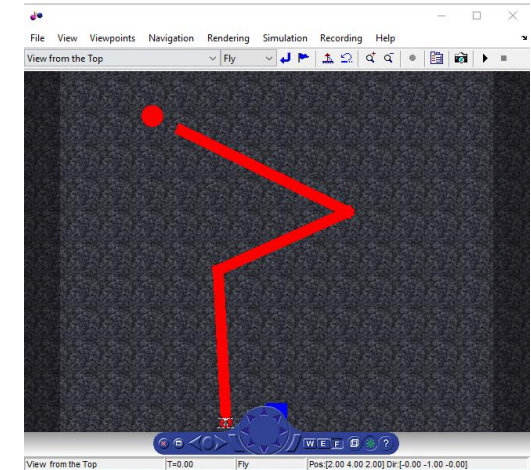
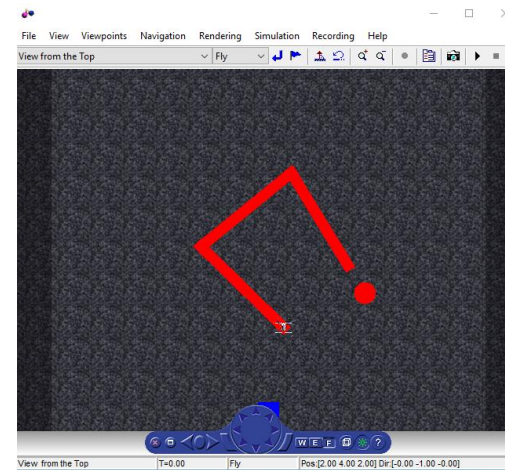
Problem Statement: Design an autonomous line follower with landing on the circle



Things to keep in mind!

Simulations

- Test on different tracks
- Use Track Builder App



Things to keep in mind!

Simulations

- Do not use computationally expensive algorithms
 - It is a toy hardware
- Do not transfer complete images between Image Processing and Path Planning
- Test your algorithm with different tracks
- Have lots of fun 😊



Preview to Day 2 & 3



Day-wise Agenda

Day 2

START TIME (EST)	END TIME (EST)	Day 2 23 June 2025
9:00	9:15	Welcome back and recap
9:15	11:00	Working on simulations
		Practice Round 1
11:00	11:15	Team 1
11:15	11:30	Team 2
11:30	11:45	Team 3
11:45	12:00	Team 4
12:00	13:00	<i>Lunch Break</i>
		Practice Round 1
13:00	13:15	Team 5
13:15	13:30	Team 6
13:30	13:45	Team 7
13:45	14:00	Team 8
		Practice Round 2
14:00	14:15	Team 1
14:15	14:30	Team 2
14:30	14:45	Team 3
14:45	15:00	Team 4
15:00	15:15	Team 5
15:15	15:30	Team 6
15:30	15:45	Team 7
15:45	16:00	Team 8
		<i>Day End</i>



Practice Rounds

- To get acquainted with the hardware
- Calibrate and test algorithms
- 3 slots of 15 mins
- Ask us questions



Day-wise Agenda

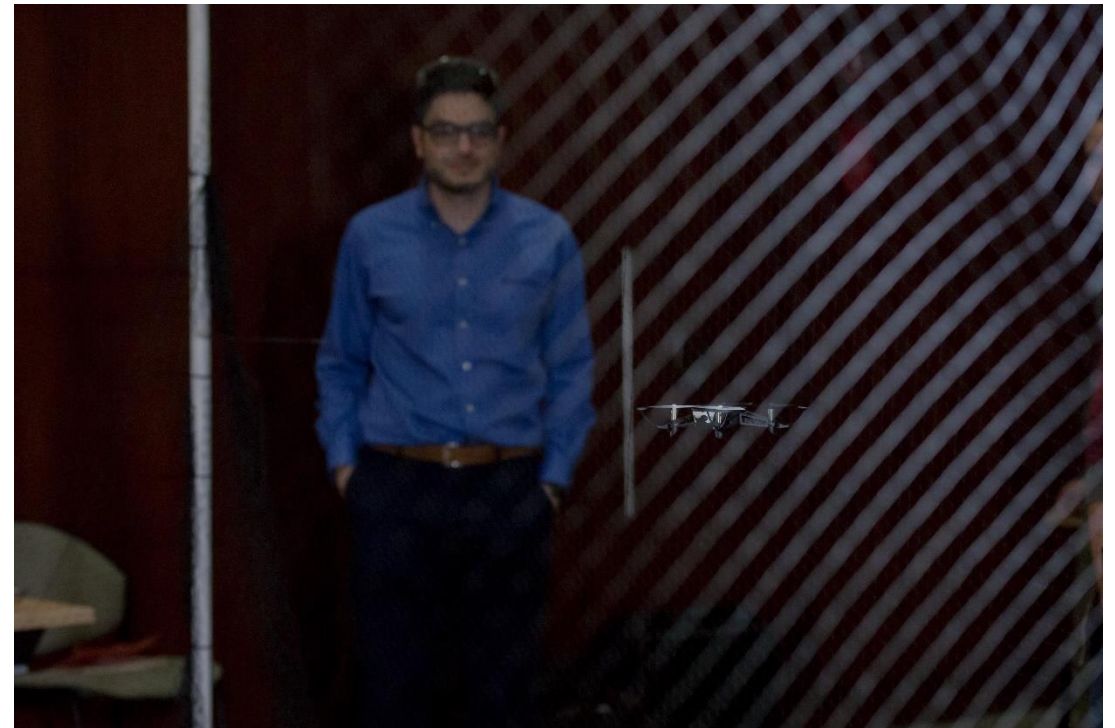
Day 3

START TIME (EST)	END TIME (EST)	Day 3 24 June 2025
9:00	9:15	Welcome back and recap
9:15	10:00	Hover the drone to test the hardware (5 mins per team)
		Optional Practice Round 3
10:00	10:15	Team 1
10:15	10:30	Team 2
10:30	10:45	Team 3
10:45	11:00	Team 4
11:00	11:15	Team 5
11:15	11:30	Team 6
11:30	11:45	Team 7
11:45	12:00	Team 8
12:00	13:00	<i>Lunch Break</i>
		Final Competition
13:00	13:15	Team 1
13:15	13:30	Team 2
13:30	13:45	Team 3
13:45	14:00	Team 4
14:00	14:15	Team 5
14:15	14:30	Team 6
14:30	14:45	Team 7
14:45	15:00	Team 8
15:00	15:30	<i>Certificates, Interaction, feedback, and giveaways</i>
		<i>Day End</i>



Live Round

- One 15 min slot
 - 7 chances to run the hardware
 - Nominate the flight to be evaluated
- Graded based on number of track sections completed
- Judges will decide the score
- No help from us



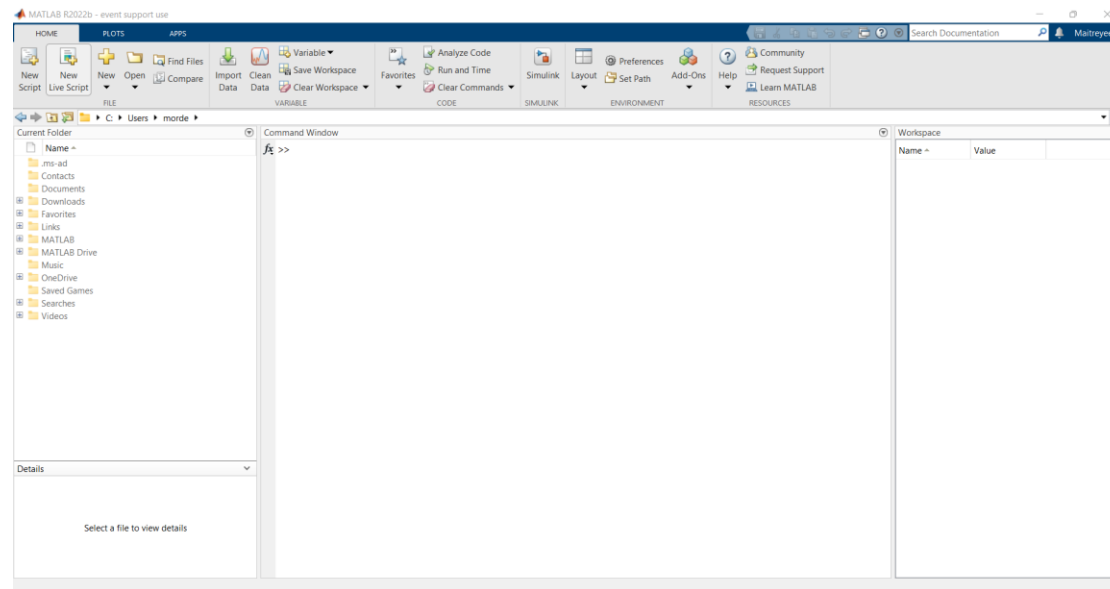
[Activity] Hardware Deployment Setup



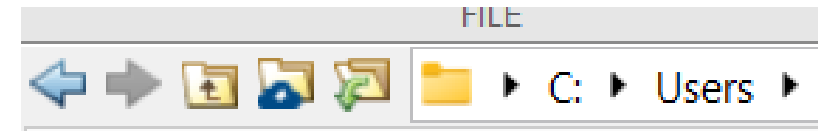
Steps to configure your MATLAB Drive

One team member ONLY - Windows

STEP 1: Open MATLAB R2022b



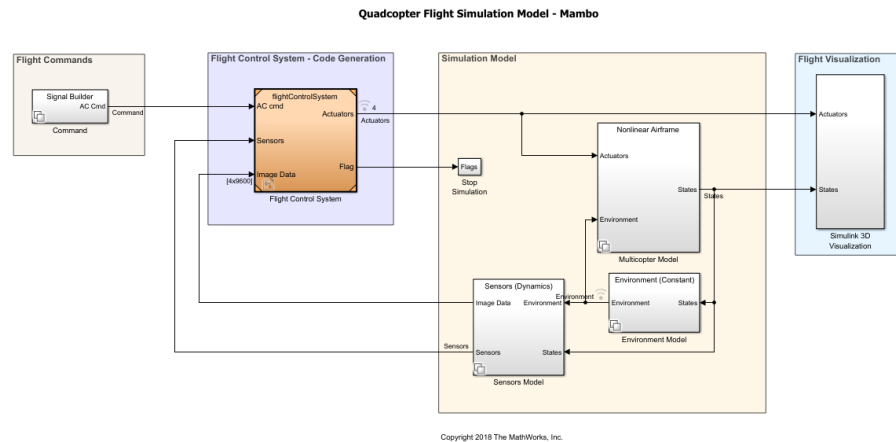
STEP 2: Install MATLAB Drive Connector



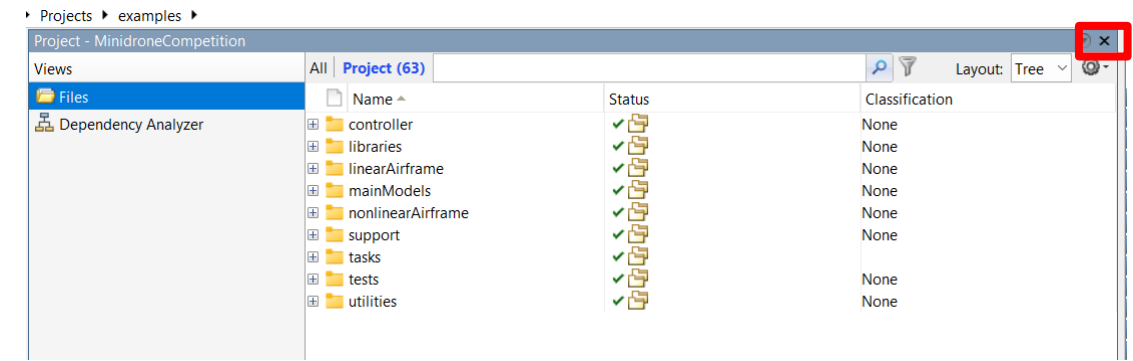
Steps to configure your MATLAB Drive

One team member ONLY - Windows

STEP 3: Open the Simulink Model



STEP 4 & 5: Change Directory and close the project

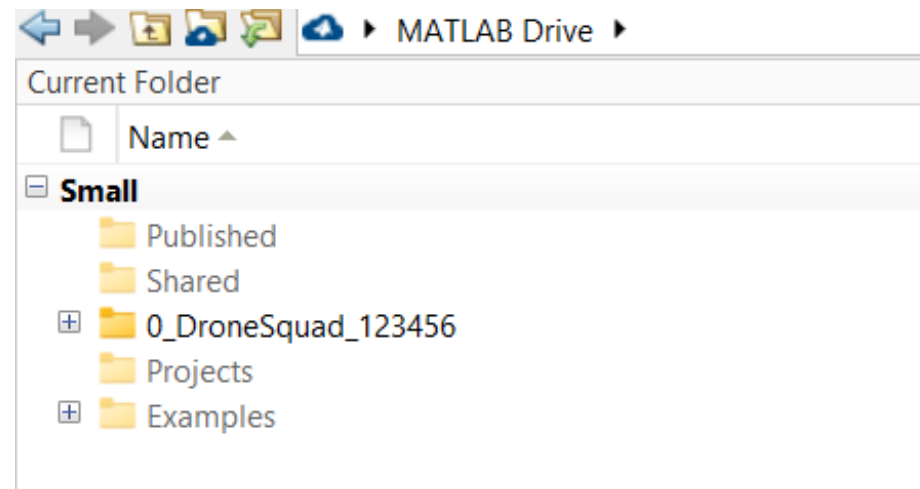


```
>> parrotMinidroneCompetitionStart
```

Steps to configure your MATLAB Drive

One team member ONLY - Windows

STEP 6 & 7: Edit project name and copy to your MATLAB Drive



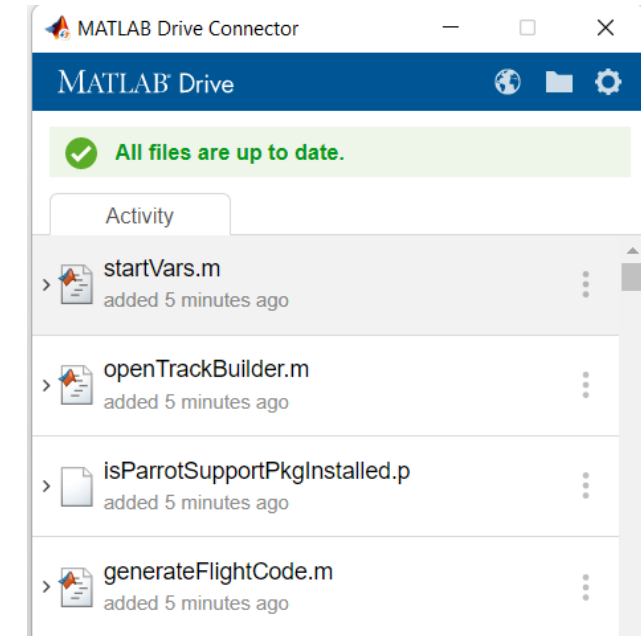
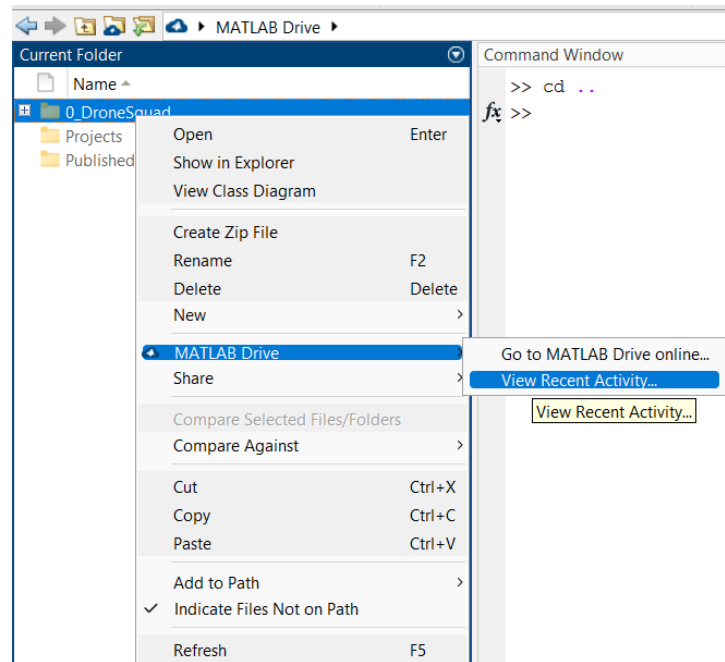
<TeamSeqNo_TeamName_HardwareNo>



Steps to configure your MATLAB Drive

One team member ONLY - Windows

- STEP 8 : View Recent Activity
- STEP 9: Ensure it is up to date



Steps to configure your MATLAB Drive

One team member ONLY - Windows

STEP 10: Share the renamed folder to **<MWTechnicalLead E-mail ID>** with '**Can Edit**' permissions

Shared With			
Me	Owner	Accepted	
	Can Edit	Accepted	✕

STEP 11: Wait for Acceptance

Things to keep in mind!

Deployment

- Get the right colors using the color thresholder app
- Do not use computationally expensive algorithms
 - It is a toy hardware
- Connect to your Minidrone only!
 - No flying outside the arena
- Test your algorithm in simulations with different tracks
- Have lots of fun 😊





MathWorks Minidrone Competitions

Learn Model-Based Design. Show off your skills. Have fun!



Scan the
QR code to
indicate your
interest

